

An NLP-Based Complaint Classification System for Barangay Reporting with Human-in-the-Loop Learning

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Lawrence B. Gonzaga
Warren Dominic S. Badong

ATENEO DE DAVAO UNIVERSITY
SCHOOL OF ARTS AND SCIENCES
COMPUTER STUDIES CLUSTER

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Abstract

This study develops a multilingual three-class text-classification component for a proposed barangay complaint-reporting system. A controlled synthetic research corpus of 24,000 narratives covers 60 Valid complaint topics, 28 Suspicious clarification reasons, 30 Spam submission types, and four assigned language groups. The 118 categories guide scenario construction and auditing, rather than serving as prediction classes. A TF–IDF and linear support vector machine (SVM) is compared with a Multinomial Naive Bayes baseline. A three-fold scenario-grouped grid search evaluated 18 SVM configurations. A separate five-fold grouped cross-validation on the training partition yielded mean Macro-F1 scores of 1.0000 for SVM and 0.99245 for Naive Bayes, with standard deviations of 0.00000 and 0.00097. On 2,399 validation records, SVM achieved 1.0000 Macro-F1 while Naive Bayes achieved 0.9921. The previously evaluated 2,388-record synthetic test partition yielded Macro-F1 scores of 1.0000 and 0.9916, respectively; these scores do not establish authentic-report generalization. Training-size learning curves, 15-epoch auxiliary stochastic-gradient hinge-loss curves, class-wise confusion matrices, language-group evaluation and feature-ablation diagnostics help interpret these scores. These observations characterize performance on researcher-generated messages and cannot independently establish reliability on previously unseen styles of citizen reporting. Subsequent reproducibility verification corroborated the original recorded metrics on the unchanged corpus. A separate meaning-preserving word/character candidate, with and without grouped sigmoid calibration, tied the reference at 1.0000 development validation Macro-F1; no classification improvement was demonstrated and no candidate test evaluation was performed. Probability reliability outside the synthetic development distribution, administrator verification, accountable case management and controlled offline feedback require independent validation and end-to-end implementation testing.

Keywords: barangay reporting; complaint classification; natural language processing; TF–IDF; support vector machine; controlled synthetic data; group cross-validation; human-in-the-loop.

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Chapter 1

Introduction

1.1 Background of the Study

Citizen complaint management is an essential function of barangay governance in the Philippines, where residents report concerns such as public safety issues, infrastructure problems, and community disturbances. Despite its importance, many barangay offices continue to rely on manual validation processes in handling complaints, resulting in delays, inconsistencies in evaluation, and inefficiencies in overall service delivery.

Davao City Reports offers a relevant local precedent for online submission, tracking references, status history, complaint routing and resident feedback. However, a municipal platform is not evidence that an identical barangay application has already been deployed or validated [7]. The proposed study therefore separates case-management design from the experimentally evaluated classification component. It addresses the need for traceable administrative records while examining whether sparse-text NLP models can distinguish coherent complaints, materially unclear complaints and clearly non-complaint submissions.

Filipino complaint-writing research and synthetic-data evaluation studies motivate sensitivity to informal expressions, code-switching and generation artifacts [8, 35, 28]. The text-only TF–IDF plus Linear SVM prototype reported here can be compared with a Naive Bayes baseline on the controlled corpus, but operational benefits require authentic external evaluation. Administrator review, measured annotation quality, controlled retraining and accountable case tracking are included as proposed extensions that must be implemented before their effectiveness can be claimed.

1.2 Statement of the Problem

This study aims to develop an automated barangay complaint classification system using Natural Language Processing (NLP) and machine learning to improve the accuracy, efficiency, and reliability of complaint processing.

Specifically, the study seeks to answer the following questions:

1. How can a web-based barangay complaint system be developed to support efficient submission, processing, and lifecycle tracking of citizen complaints?
2. How can Natural Language Processing (NLP) techniques using TF-IDF and Support Vector Machine (SVM) classify complaints into valid, suspicious, and spam categories?
3. How can a Human-in-the-Loop (HITL) mechanism be integrated to provide traceable staff corrections and evaluate whether controlled retraining improves the classification model?
4. How effective is the proposed system in terms of accuracy, precision, recall, and F1-score in classifying complaints compared to baseline or traditional methods?

1.3 Objectives of the Study

This study aims to develop and evaluate an automated barangay complaint classification system using Natural Language Processing (NLP) and machine learning.

Specifically, the study aims to:

1. Develop a web-based barangay complaint system that streamlines the submission, processing, and lifecycle tracking of citizen complaints.
2. Evaluate the effectiveness of TF-IDF and Support Vector Machine (SVM) in classifying multilingual complaint text (Bisaya, Filipino, and English) into valid, suspicious, and spam categories.
3. Design and evaluate a Human-in-the-Loop (HITL) protocol for administrator review, correction logging, curated feedback and controlled model updates.
4. Evaluate the performance of the proposed system in terms of accuracy, precision, recall, F1-score, with subgroup, boundary-case and held-out challenge tests; evaluate operational workload separately when instrumented.

1.4 Significance of the Study

This study contributes to the advancement of intelligent e-governance systems by improving barangay-level complaint management through automated text classification. It addresses inefficiencies in manual validation and supports more efficient and data-driven public service processes.

From a technical perspective, this research contributes to Natural Language Processing (NLP) and machine learning classification by applying these methods in a multilingual context involving Bisaya, Filipino, and English. It also extends existing work by integrating a Human-in-the-Loop (HITL) mechanism, proposing a testable route for administrator-verified feedback and controlled future model improvement.

The study introduces a framework that combines NLP-based text classification with a state-transition workflow to ensure structured tracking of complaint lifecycles. This

is intended to support accountability and transparency through traceable administrative actions; operational gains have not yet been demonstrated.

For stakeholders, the system provides practical benefits to barangay officials by potentially improving the traceability of administrative decisions; reductions in manual workload and processing time require separate field evaluation.

Overall, the study bridges the gap between traditional e-governance systems and adaptive intelligent solutions, offering a scalable approach that can be adopted by other local government units to improve administrative efficiency and transparency.

1.5 Scope and Limitations

Scope

This study focuses on a proposed web-based barangay reporting prototype built with a Next.js/TypeScript citizen and administrator interface, a FastAPI/Python backend, and PostgreSQL persistence. The backend incorporates Natural Language Processing (NLP), Term Frequency–Inverse Document Frequency (TF-IDF), and a linear Support Vector Machine (SVM) for recorded classification of Bisaya, Filipino, English, and mixed/code-switched complaint inputs as Valid, Suspicious, or Spam. The full-stack selection is an implementation design; the synthetic corpus and EDA do not alone demonstrate that every application feature is deployed.

The intended system incorporates a Human-in-the-Loop (HITL) mechanism for staff verification and recorded corrections, with controlled later model retraining. PostgreSQL stores complaint data, review history, and distinct workflow states that track each complaint from submission to resolution.

Limitations

This study uses a researcher-generated synthetic complaint dataset and does not include full-scale real-time deployment across multiple barangays. The system is restricted to text-based complaints and does not support multimedia inputs such as images, videos, or audio.

Additionally, the system does not include advanced user identity verification and cannot fully guarantee the authenticity of anonymous submissions. The model's performance may vary depending on dataset size, data quality, and the level of code-switching in multilingual inputs. Lastly, the system functions only as a decision-support tool and does not replace the final judgment of barangay officials.

1.6 Research Questions for Controlled Data Adaptation

Beyond interface digitization, the experimental contribution concerns the effectiveness of a controlled synthetic corpus. The study asks whether explicitly varying linguistic

form, incident context, and ambiguous-versus-meaningful intent provides a reproducible means of testing a three-class complaint classifier. A second question asks whether class-balanced internal scores are stable on independently written challenges rather than merely recognizing the generator's recurrent lexical patterns. A third question concerns how reviewed administrator corrections could be introduced without contaminating held-out evaluations or silently changing live predictions. These are evaluated or scoped separately: dataset construction and baseline model testing have experimental results; operational improvements and feedback-driven retraining remain proposed until independently tested.

1.7 Research Status and Delimitations

The completed experimental corpus comprises 24,000 researcher-constructed narratives partitioned for model development. The study does not claim a representative sample of Davao barangay submissions, independent human adjudication of all labels, comprehensive language fluency certification, or live deployment. The 118-item category register is a *scenario and auditing framework*; it is not the target output space of the classifier. The deployment interface would predict only Valid, Suspicious, or Spam, with an administrator retaining final authority. Accuracy alone is insufficient to establish deployment readiness because authentication, privacy, referral, latency, state transition, risk escalation and model-update controls require separate tests [15, 18].

Chapter 2

Review of Related Works

This chapter presents a review of existing literature organized thematically to identify the gaps that this study addresses. The themes cover barangay complaint management systems, NLP and machine learning for text classification, Human-in-the-Loop and adaptive learning systems, and workflow and lifecycle management.

2.1 Barangay Complaint Management Systems

Barangay reporting involves community concerns whose administrative handling is distinct from automatic assessment of narrative text. The Local Government Code provides a general local-governance context, while Davao City Reports and the corresponding City Government Citizen's Charter document tracking references, routing and administrative feedback mechanisms [1, 7]. These sources establish relevant local design precedents; they do not establish that every barangay currently has the same procedures, or that an AI classifier has already improved staff response times. The present study proposes a barangay-specific text triage and auditable case-management design that would require local approval and user testing before operational claims could be made.

2.2 NLP and Machine Learning for Text Classification

The proposed operational model would convert resident-authored narratives; the recorded research instead converts researcher-constructed synthetic narratives into sparse numerical features using TF–IDF, then estimates a researcher-defined text-suitability class using a linear SVM. The main reported study uses word-unigram TF–IDF with the SVM and a separate Naive Bayes baseline; a character-feature ablation is also reported. Word-plus-character fusion and sigmoid probability calibration were separately evaluated on existing development validation, tying the reference Macro-F1. They are not the original reference feature configuration or a validated deployed model. Scikit-learn documentation describes the chosen vectorizer, estimator and calibration mechanisms [18, 43, 44].

Philippine complaint pragmatics and code-switching motivate retaining informative words, nonstandard spelling and context rather than equating short or emotional reporting

with Spam or incompleteness. De Leon and Parina document variations in Filipino complaint realization, while Bender and Friedman emphasize documenting the language and speaker contexts represented by NLP datasets [8, 28]. These studies do not verify the four assigned language groups in the generated corpus. Synthetic-data literature further warns that generated examples can introduce patterns that inflate performance under the same generator distribution [35, 14].

2.3 Human-in-the-Loop and Adaptive Learning Systems

Human-in-the-Loop (HITL) systems incorporate human assessment into model-assisted decisions. Prescriptive annotation guidelines and reviewer disagreement are crucial when class definitions require judgments about narrative completeness [33, 34]. Cost-sensitive filtering and deferral literature further motivate human verification of potentially consequential Spam predictions [37, 38].

Feedback collection must not be confused with completed active learning. A process that accumulates administrator-verified labels and periodically refits a candidate classifier is more precisely described as *human-feedback-driven batch retraining*. The stronger active-learning term applies when informative samples are deliberately selected for annotation and that sampling procedure is evaluated. Research finds that active-learning gains depend on dataset, model and sampling design; the present study therefore treats the feedback policy as a proposal to validate [41, 40].

2.4 Workflow and Lifecycle Management

Case management is an administrative state-transition process, not a fourth or fifth prediction class. Accepted cases can be Unresolved, Ongoing and Resolved; Overdue, Referred and Reopened are logged events rather than text-classifier outputs. Davao City Reports presents a local precedent for history, status tracking and resident follow-up [7]. Republic Act No. 6713 and the Anti-Red Tape Authority's explanation of RA 11032 support traceable communications and context-sensitive service processing rules; these should not be misread as universal guarantees that every incident is fully resolved within one fixed interval [29, 30]. A proposed workflow therefore requires responsible-person assignment, logged actions, deadlines derived from applicable procedures, evidence review and a reconsideration mechanism.

2.5 Theoretical Framework

The theoretical framework of this study is anchored on six key theories and models: Natural Language Processing (NLP), Text Classification Theory, Term Frequency–Inverse Document Frequency (TF-IDF), Support Vector Machine (SVM), Human-in-the-Loop (HITL) and Active Learning, and State Transition Theory. Each theory is discussed individually below to establish the academic foundation of the proposed system.

2.5.1 Natural Language Processing (NLP)

Natural Language Processing is a field of artificial intelligence that enables computers to transform unstructured human language into structured and machine-readable data. In this study, NLP serves as the foundational layer that processes raw complaint text submitted by barangay residents, allowing it to be analyzed by machine learning models. Studies of Filipino complaint pragmatics and data documentation motivate preserving expression differences and describing speaker/language representation explicitly [8, 28].

NLP pipelines may include text normalization, tokenization and optional stemming or stopword filtering. In multilingual complaint data, indiscriminate stopword removal or stemming may erase meaningful Cebuano particles and negation. Accordingly, the study retains a conservative text representation and learns TF-IDF vocabulary within the training folds; the experimental boundary is specified in Chapter 3 rather than implying that multilingual stemming was implemented.

The selected proposed preprocessing workflow is specified separately in Section 3.10, without introducing a second overlapping figure.

2.5.2 Text Classification Theory

Text classification theory explains the process of assigning predefined categories to textual data. It operates by learning patterns from labeled training data and applying these learned patterns to unseen inputs. This involves transforming text into numerical representations and using decision-making models to determine the most appropriate category, ensuring consistent and automated evaluation.

In this study, text classification theory underpins the entire complaint categorization process. The system learns from labeled complaint samples to distinguish between valid, suspicious, and spam complaints, enabling consistent and automated classification without relying on manual judgment.

2.5.3 Term Frequency–Inverse Document Frequency (TF-IDF)

TF-IDF is a feature extraction technique that converts textual data into numerical representations based on word importance. It assigns higher weight to words that are significant within a specific document while reducing the influence of common terms that appear across many documents. TF–IDF provides a transparent sparse numerical representation; its value for this corpus is assessed empirically rather than assumed in advance [43]. In this study, TF-IDF transforms multilingual complaint text into structured numerical vectors, enabling the SVM classifier to process the data effectively.

The experimental word-vector baseline and the separately evaluated combined-feature candidate are distinguished explicitly in Section 3.10.

2.5.4 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification tasks. It works by identifying an optimal hyperplane that separates different classes in a high-dimensional feature space. SVM is well-suited for text classification because of its effectiveness with high-dimensional and sparse datasets, which are common in NLP applications. For the chosen representation and dataset, the SVM is compared directly with a Multinomial Naive Bayes baseline; the outcome is limited to the specified generated-data evaluation [18]. In this study, SVM operates on TF-IDF-generated features to classify complaints into valid, suspicious, or spam categories.

2.5.5 Human-in-the-Loop (HITL) and Active Learning

The Human-in-the-Loop (HITL) framework integrates human expertise into the machine learning lifecycle to improve model performance and reliability. It can be combined with active learning when a documented query strategy deliberately selects examples for human labeling; merely storing corrections is insufficient to demonstrate improved accuracy.

Batch active-learning research discusses sample selection and stopping conditions, but its gains cannot be assumed for this three-class corpus without a direct test [40, 41]. In the planned deployment, when barangay administrators correct system-generated classifications, the workflow would record version-specific corrections for quality review and possible future offline retraining. This governed feedback path could allow the system to adapt to local linguistic variations and behavioral patterns over time.

2.5.6 State Transition Theory

State transition theory provides the foundation for workflow management within the system. It models a process as a finite set of states and defines the transitions between them based on specific conditions or actions. In the context of this study, state transition theory governs the complaint lifecycle, defining the movement of a complaint through Unresolved, Ongoing, and Resolved states. Closure, reopening, referral and overdue flags are recorded as administrative events rather than additional prediction classes. This ensures that every complaint is systematically tracked and that barangay officials can monitor progress at each stage, promoting accountability, transparency, and proper resolution.

2.6 Barangay Complaint Domains and Legal Jurisdiction

The Local Government Code (RA 7160) establishes a broad general-welfare and administrative foundation for barangay-level services, dispute-related procedures and community coordination [1]. This foundation must not be misread as authorizing a barangay to settle every report. Solid waste concerns are informed by RA 9003; disaster hazards by RA 10121; adolescent and youth participation by RA 11768; and accounts of financial account scams by RA 12010 [2, 3, 5, 6]. Reports involving violence against women and their children may be meaningful community reports but require safeguards and appropriate authorities; RA

9262 does not make these matters ordinary conciliable barangay disputes [4]. Therefore, this work defines a Valid label to mean a *comprehensible report of a community concern*, irrespective of whether its outcome requires referral. This is a text-classification definition, not a legal finding that a report is factually true.

2.7 Complaint Pragmatics: What People Actually Write

Complaints are communicative acts, not uniform written incident reports. Study of Filipino complaints in English and Tagalog highlights variation in complaint realization and directness [8]. Research on anonymous online complaint discourse provides evidence that negative evaluation, criticism, and emotional expression may be part of complaint writing [10]. These studies motivate variation in the intended message: concise description, direct request, rhetorical question, emotional follow-up, secondhand concern and partially ordered narratives. They do not supply representative national percentages for each form. Accordingly, this project *chooses* experimental writing quotas rather than presenting them as natural reporting prevalence. A short or emotional message can be Valid; the intended meaning, not politeness, determines the classification label.

2.8 Cebuano, Filipino, English and Code-Switching

Residents may use Cebuano/Bisaya, Filipino, English and mixed forms. Porlas [9] provides a linguistic reference for grammatical patterns in Philippine language code-switching. The generator therefore uses language-associated seed expressions rather than mere random replacement of translated tokens. Some submissions are inherently language-neutral, such as nonsemantic number strings or links. Language-group values in the released dataset refer to generation allocation; they have not been confirmed by independent language annotation. This distinction is important in subgroup reporting.

2.9 Controlled Synthetic Data as an Experimental Instrument

Synthetic examples can deliberately introduce rare or challenging message forms, vary attributes systematically, and preserve provenance without relying on sensitive resident submissions. Long et al. survey synthetic generation, curation and evaluation [12]; Chim et al. investigate style and distributional comparisons for user-generated synthetic data [13]. Shaib et al. report recurring syntactic templates in generated text, a direct concern for lexical classifiers [14]. Counterfactual augmentation can be used to test decision boundaries, but its benefits are not guaranteed. Kaushik et al. describe counterfactual data construction while Huang et al. document a negative result where counterfactually augmented training did not improve generalization [16, 17]. These findings justify *measuring* improvements on a fixed and independent challenge set rather than asserting that synthetic control automatically yields a better deployed model.

2.10 Generalization, Annotation Artifacts and Behavioral Testing

Ribeiro et al. propose CheckList, an NLP testing methodology that probes capabilities using perturbations and expected behavior rather than relying exclusively on average accuracy [15]. In this project, priority probes include preserving Valid labels under spelling and politeness changes, preserving Spam labels under more natural promotional wording, and changing the class when a previously ambiguous report acquires sufficient essential information. Failure rates on independently written probes help expose missed decision boundaries. Near-perfect synthetic test performance is informative only within the assumptions and lexical distributions of that dataset.

2.11 Evaluation Design and Leakage Prevention

A correct evaluation partitions examples by underlying incident or generation group before fitting text features. A scikit-learn Pipeline places TF-IDF fitting *within* each training fold. Group-aware cross-validation avoids reusing a related scenario in both the corresponding training and validation folds. In a prospective design, an untouched test estimates performance after configuration is fixed. In this study the original synthetic test has already been evaluated; existing validation supports development diagnostics, and later verification is not independent new testing. A separately authored challenge set would probe distribution shift, but no missing challenge input was regenerated in this work. Optimizing against the challenge repeatedly would convert it into another validation set [18]. `LinearSVC` does not publish per-epoch loss; a separate `SGDClassifier` hinge-loss diagnostic may illustrate convergence but must be clearly attributed to that auxiliary estimator.

2.12 Human-in-the-Loop Governance and Operational Readiness

Amershi et al. describe the importance of human participation in interactive machine learning [19]. For barangay complaints, reviewed corrections must be traceable, vetted for mislabeling and sensitive information, and evaluated offline before promotion to production. The approach treats the machine's prediction as provisional. A deployed tool additionally needs authenticated reviewer access, audit retention rules, complaint referrals, error handling, disaster and safeguarding escalation, workflow checks, performance monitoring, and rollback of problematic model versions. These requirements are independent of training dataset size.

2.13 Data-Centric Validation and the Limits of Synthetic Benchmarks

The study treats synthetic narratives as a controlled experimental instrument rather than as a statistically representative sample of barangay residents. This position requires both a defense and a limitation. Long et al. [12] organize synthetic-data research around generation, curation, and evaluation rather than treating large-scale generation as sufficient evidence of quality. Chim et al. [13] emphasize examining the differences between source text and generated text, including style and downstream usefulness. In this study, repeatable generation makes it possible to vary language, issue, and ambiguity by design, but repeatability does not automatically ensure semantic accuracy or transfer to unfamiliar writing.

The possibility of *annotation artifacts* is particularly important. Gururangan et al. [24] demonstrated that superficial regularities introduced by dataset construction can support shortcuts in a natural-language task. Shaib et al. [14] study recurring syntactic templates in generated text. These findings motivate three distinct checks in the present research: (a) strict non-overlap of recorded scenario-family identifiers between partitions; (b) measurement of text-length and lexical proxies for label identity; and (c) linguistic tests beyond the generator-derived validation distribution. Group separation prevents the same registered scenario family from occurring in both sides of a split, but it does not eliminate shared words or class-specific narrative frames produced by the same generator.

2.14 Why Cross-Validation and a Separate Validation Set Both Matter

A single train–test split can yield a score that depends on which scenarios happened to be selected for evaluation. Cross-validation repeats training across multiple held-out subsets and therefore exposes fold-to-fold variation. However, ordinary K-fold splitting allows closely related variations of one generating scenario to appear on both sides. The study instead employs `StratifiedGroupKFold`, which attempts to maintain class proportions while ensuring that each group appears in only one held-out fold [25]. Importantly, the TF-IDF vectorizer is inside the scikit-learn pipeline; its document frequencies and vocabulary are learned from the training portion of each fold, never from the held-out fold [23].

The three-fold grid search is used for *selection* of the SVM parameters. A subsequent five-fold group-based analysis is reported as a stability diagnostic for the selected SVM and the fixed Naive Bayes baseline. The fixed validation partition is then assessed separately. These measures address different questions: the grid compares alternatives; the second set of folds inspects repeated partition behavior; and validation measures performance on the designated sample that was not used to fit model weights. The reserved final test partition is *not* used for the present study tuning or the reported validation experiments.

2.15 Behavioral Testing Beyond Aggregate Accuracy

Ribeiro et al. [15] argue that aggregate accuracy can miss systematic linguistic weaknesses and present the CheckList approach for behavioral tests such as changes that should preserve or alter a prediction. This suggests that the final application needs controlled cases involving paraphrases, negation, incomplete references, code-switching, and urgent but informal reports. The present validation analysis examines assigned-language subgroups and simple surface-feature baselines; it does not claim that a complete, independently adjudicated behavioral benchmark has been established. The distinction is necessary to avoid presenting promising model experiments as production validation.

2.16 Documentation and Independent Annotation

Gebru et al. [20] propose datasheets that describe the origin, composition, intended uses, and limitations of datasets. We adapt this principle by retaining metadata for each generated example: source identifiers, classification rationale, intended language, scenario-family membership, generation method, and partition. Artstein and Poesio [21] explain why annotation agreement must be interpreted in context rather than treated as interchangeable with classifier accuracy. A future blind review with independent raters is therefore a separate requirement, and the present records are explicitly flagged as not yet human-reviewed. These caveats are reflected in the scope and interpretation of Chapter 4.

Chapter 3

Methodology

3.1 Conceptual Framework

The study has two mutually connected, but experimentally distinct, phases. Phase I develops and evaluates a controlled text-classification artifact without requiring sensitive resident complaints. Phase II describes how that artifact would support a human-governed barangay reporting system. The separation is essential: achieving a test-set score completes a model experiment, whereas operating a public-facing reporting system requires application, governance and safety verification. Figure 1 presents these phases with an explicit release gate and a feedback loop that returns reviewed corrections to *offline* retraining instead of changing a live classifier automatically.

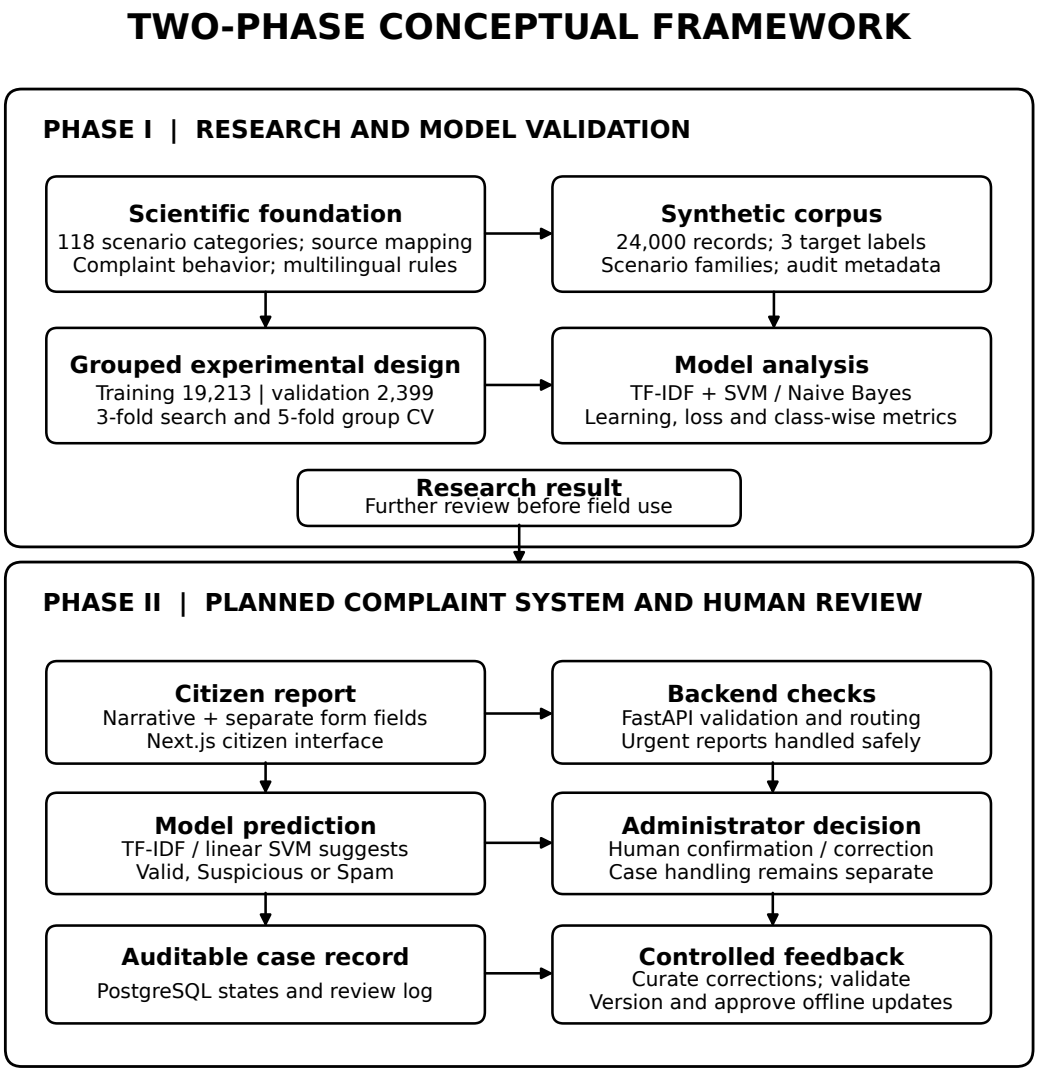


Figure 1: Expanded two-phase conceptual framework separating research preparation, model selection, provisional inference, human review and controlled feedback. Arrows distinguish the measured research workflow from the proposed operational workflow; administrator feedback is reviewed offline rather than automatically deployed.

3.1.1 Phase I: Evidence-Based Corpus Development

The offline phase begins with a source register: national legislation provides examples of community responsibility and safeguards, local reporting guidance describes information useful for incident review, linguistic studies inform complaint speech acts and code-switching, and synthetic-data research informs variation and evaluation controls. These sources *inform* the scenario design. They do not certify each generated sentence as authentic or prove that the proportions are representative. A 118-item researcher-defined

taxonomy supplies 60 Valid issue topics, 28 Suspicious clarification reasons and 30 Spam types. Each item receives an explicit scope definition, an intended three-class label, and references to the corresponding evidence type.

The generation stage distinguishes *scenario meaning* from *surface form*. A hypothetical clogged-drain concern can be written as an informal question or a detailed narrative without changing its class. A confusing report may instead refer only to "the thing I reported before" with no recoverable subject. A promotional solicitation lacks a community grievance altogether. Generator logic chooses a controlled language group, communication form, narrative length and contextual cues, then constructs the sentence. Every output carries source traceability, seed, topic/review/spam identifier and a scenario family. These metadata fields are excluded from the text-only classifier. Figure 2 details this process.

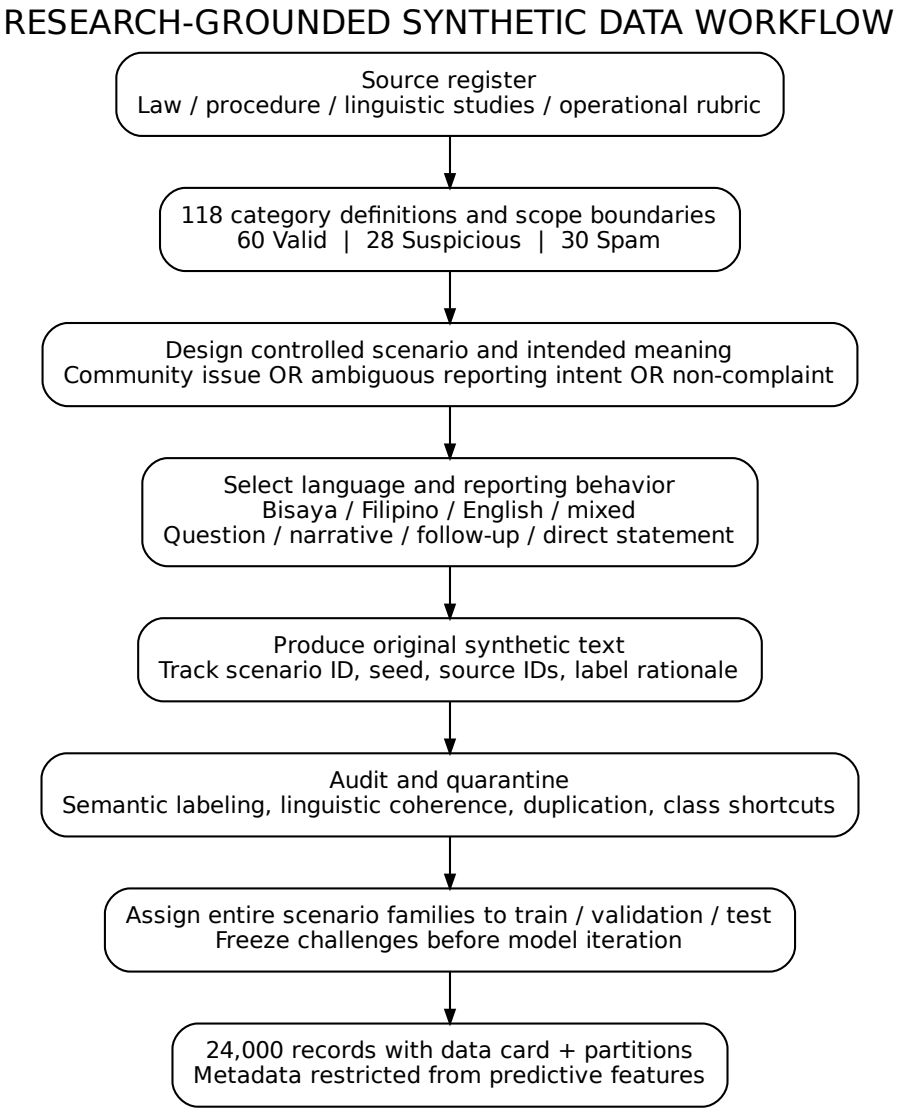


Figure 2: Scenario-first data construction and audit pipeline. The label is based on intended communicative meaning, not on incidental length, grammar, sentiment, or language.

3.1.2 Phase II: Proposed Public Reporting Service

The proposed operational system separates resident input, inference, administrative review, case management, and controlled model maintenance. A resident-facing form collects a required Subject and Complaint Details narrative; incident date/time and Purok/Sitio or an identifying landmark are supporting contextual fields when known. Geographic maps, GPS capture and photographic uploads are outside the selected core thesis scope.

The application concept uses an authenticated resident interface and an authorized staff interface; the suggested software stack is Next.js/TypeScript, FastAPI and PostgreSQL, but implementation claims require working source code and testing. No completed user-interface or government-office deployment evaluation is included in the supplied experimental evidence.

The class-prediction output is separate from the administrative disposition. Predicted Valid goes to the accepted-case workflow under the selected routing rule; predicted Suspicious or Spam goes to administrator review. Acceptance, clarification, and rejection are decisions about the submission, not automatic replacements for the text label. Proposed controls include resident status tracking, staff action logs, deadline monitoring, referrals, verification before resolution, and a reconsideration mechanism. Detailed proposed figures and the experimental–operational boundary are presented later in this chapter.

3.2 Descriptive Analysis of the Dataset

This section updates the study's earlier data-description protocol to reflect the corpus actually used for the reported experiments. The study originally considered de-identified administrative archives, subject to access approvals, and supplemental simulated examples. Those were possible acquisition pathways, not completed observations. The experimental data currently consist entirely of *researcher-constructed synthetic complaint narratives*; no authentic barangay records are presented as training data. The objective of using controlled synthetic text is to build an auditable and systematically varied classification benchmark, not to estimate how frequently a complaint type occurs in Davao City. Following Chicco et al. [27] and Bender and Friedman [28], the data's provenance, sampling choices, missing coverage and preprocessing choices are reported alongside model scores.

3.2.1 Data Sources, Construction and Composition

Source acquisition in this study refers to researching the *foundations of scenario design*: Philippine barangay governance and topic-specific laws, administrative reporting guidance, studies of Filipino complaint speech acts and code-switching, and published approaches to spam and synthetic-text validation. Each of 60 Valid topics, 28 Suspicious review reasons and 30 Spam forms is tracked in the accompanying source-to-category matrix. These references provide topic or linguistic justification; they neither certify individual generated incidents as real nor imply the 118 categories are official government labels. Generation was carried out from researcher-designed incident descriptions and style frames; the existing fixed corpus is not being regenerated in this revision.

The corpus contains **24,000** records and **21 recorded fields**. Of those, the predictive input is `complaint_text` and the target is `classification`. Supporting columns record category, language assignment, scenario or template family, provenance, assigned partition, and review-status flags. The raw unmodified CSV remains preserved within the research repository. All 24,000 source-type entries are marked `synthetic_researcher_constructed_unreviewed`. This label distinguishes the corpus from real archived reports and from independently annotated gold-standard data.

Table 1: Observed corpus composition and provenance

Dimension	**Records**	**Interpretation**
Entire research corpus	24,000	100% researcher-generated synthetic text
Valid	8,000	33.33% of corpus
Suspicious	8,000	33.33% of corpus
Spam	8,000	33.33% of corpus
Bisaya/Cebuano (assigned)	6,000	25% of corpus
Filipino (assigned)	6,000	25% of corpus
English (assigned)	6,000	25% of corpus
Mixed/code-switched (assigned)	6,000	25% of corpus
Verified human annotations	0	No independent label review recorded
Authentic archived complaints	0	Not a sampled administrative archive

3.2.2 Exploratory Data Analysis and Text Characteristics

Exploratory Data Analysis (EDA) was computed on the existing unprocessed `complaint_text` values using whitespace-delimited word counts, class totals, assigned language totals and duplicate checks. Token length in this table is a reproducible whitespace-word count, not a universal measure of syllables or morphologically segmented Cebuano vocabulary. The corpus has no missing complaint narrative and no exact duplicate strings, but neither condition rules out meaning-equivalent paraphrases, repeated phrasal openings or near-duplicate source patterns.

Table 2: Measured narrative lengths by supervisory label

Label	n	**Mean**	**Median**	**Minimum**	**Maximum**
Valid	8,000	20.76	20	5	48
Suspicious	8,000	29.78	28	8	66
Spam	8,000	14.90	15	1	35

The overall corpus has a mean length of 21.81 words, median of 21 words, and observed range of 1–66 words. Figure 3 displays the unequal within-class distributions. Suspicious entries average substantially longer than Spam, showing that an initially balanced class design can still encode unintended class cues in narrative length. This motivates the length-only and character-ngram diagnostics reported in Chapter 4; equal class totals alone cannot prevent superficial shortcuts.

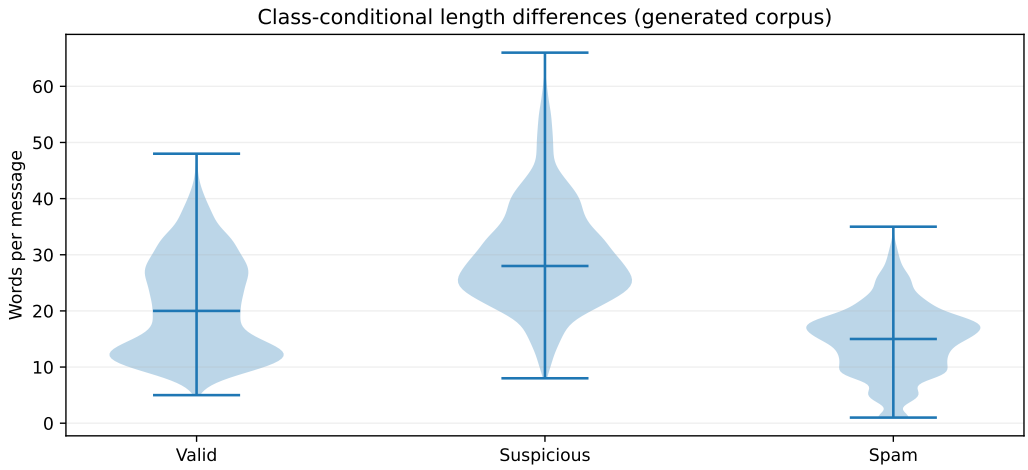


Figure 3: Class-wise generated narrative length distributions, showing that equal class counts do not imply equal wording characteristics.

The data were organized into 19,213 training, 2,399 validation and 2,388 test entries (80.05%, 10.00% and 9.95%, respectively after rounding). Small deviations from a nominal 80/10/10 allocation arise from assigning shared scenario-family IDs to the same partition. This actual partition replaces the earlier *proposed* stratified 80/20 train–test arrangement; neither design should be simultaneously presented as implemented. Neither the assigned language proportions nor the target-class proportions are estimates of demographic or reporting-channel prevalence.

3.2.3 Inherent Biases, Sampling Gaps and Residual Risks

Synthetic construction eliminates the need to treat past administrative acceptance or rejection decisions as ground truth, but introduces different biases. *Selection bias* arises because researchers choose the 118 scenarios, seed incidents and forms of ambiguity, which may omit types of future complaints. *Generator-pattern bias* arises from repeated frames, vocabulary and length conventions. *Language coverage bias* arises when Bisaya and mixed-language expressions are assigned evenly without verifying natural idiom or dialect variation. *Annotation bias* arises because labels are applied by rule and no independent annotator agreement has yet been measured. *Operational distribution shift* arises because real inputs could have different class base rates, spelling mistakes, topics, contextual references or user behaviors. *Temporal drift* could affect slang, complaint topics and spam strategies after generation.

Unlike studies based on real administrative archives, this particular corpus cannot directly measure historical discrimination or exclusion *within actual office records*, because no such records were sampled. Nevertheless, if the system later incorporates staff corrections, differences in administrator practice, submission channels, access to the internet and language preference could be reproduced through feedback. These risks must be audited separately rather than claimed as eliminated. Documentation of intended linguistic coverage and limitations follows the data-statement rationale in [28].

Table 3: Bias risks, recorded controls and remaining gaps

Risk	**Control already reflected in corpus / experiments**	**Residual validation need**
Class imbalance	Deliberate 8,000/8,000/8,000 allocation	Test differing real-world base rates
Language imbalance	Four equal generator-assigned groups	Fluent-speaker and sub-language review
Labeling subjectivity	Written three-class rubric, 118 source-linked reasons/topics	Blind dual annotation, Cohen's kappa
Exact duplication	No exact repeated narrative among 24,000 records	Check near duplicates, shared seed syntax
Partition leakage	Scenario-family group separation; fold-only TF-IDF fitting	Test unrelated root-seed and writing-family holdouts
Generator shortcuts	Length and feature-ablation diagnostics	More adversarial minimal-pair tests
Administrative bias	No past office disposition field used as model input	Audit reviewer corrections before retraining

3.2.4 Cleaning, Preprocessing and NLP Representation

The recorded pipeline uses the narrative as the *sole predictive feature*. Basic import and integrity checks ensure required text is present and encoding can be read. Conservative normalization handles whitespace and case through the TF-IDF analyzer. Word tokenization and configurable unigram/bigram extraction construct sparse numeric vectors. The `TfidfVectorizer` is fitted *inside each training fold* and then applied with `transform` to held-out data; fitting it before splitting would leak corpus vocabulary and term frequencies. Dataset category IDs, language assignments, complaint status and reported resolution are not used as predictive features. The distinct proposed preprocessing workflow is described in the operational-workflow section; the completed experiments use word TF–IDF rather than the combined word–character proposal.

The earlier methodology proposed stopword filtering, stemming and lemmatization as possible steps. None is asserted to have improved results or to have been deployed in the current final TF-IDF–SVM model: deleting Cebuano negation, particles or borrowed terms could remove meaning. Likewise, named-entity masking is a privacy measure for future user-submitted text, but the present corpus consists of fictional scenarios and does not constitute a verified de-identification evaluation. The project preserves raw text for auditability and fitted preprocessing objects for consistent later inference.

3.2.5 Class Balancing, Curation and Overfitting Controls

Because the recorded study set is *already balanced by construction*, no minority-class oversampling or synthetic minority oversampling technique (SMOTE) was needed to create parity across Valid, Suspicious and Spam. The data-generation process itself is the curation/balancing intervention. SMOTE was *not applied*: interpolation between high-dimensional sparse TF-IDF vectors might not correspond to interpretable complaint messages. Likewise, class-weighted SVM training and random oversampling were discussed

as possible controls for future imbalanced feedback datasets, but should not be described as completed current experiments unless supported by run configurations. When comparing such methods in future, balancing must be restricted to the training data *inside* validation folds, never applied to held-out validation or test samples.

Regularization is the model-appropriate overfitting control: the SVM penalty parameter C was explicitly tuned over $\{0.1, 1, 10\}$. The TF-IDF n-gram range and minimum document frequency were also tuned as feature-complexity controls, with an upper document-frequency restriction and sublinear term-frequency weighting held fixed. Model selection used three-fold scenario-grouped cross-validation, supplemented with five-fold group-disjoint fixed-model diagnostics on the training partition; fixed validation performance and loss/learning curves are reported in Chapter 4. A linear SVM has no dropout layers, and the selected `LinearSVC` does not provide a neural-network-style early-stopping history. A separate SGD-based linear hinge-loss estimator supplied the reported epoch-level *auxiliary* loss curves. Regularization and grouped validation reduce some forms of fit-related risk but cannot certify transfer to unseen generator root patterns.

3.2.6 Human Annotation and Controlled Feedback

The earlier research protocol proposed that Gonzaga and Badong independently annotate a calibration subset, reconcile disagreements and calculate Cohen's kappa [21]. This remains an *uncompleted validation procedure*; the corpus marks all records `human_label_reviewed=No`. The implemented annotation method is generator-rule labeling, not independently measured consensus. A future review sample should be stratified by all three classes, language groups, category families, length bands and deliberately ambiguous boundary cases. Reviewers should receive complaint text alone, without metadata revealing the intended label, and adjudication outcomes should be preserved.

The intended Human-in-the-Loop process similarly requires an administrator to confirm or correct a provisional prediction, with decisions stored for later audit. Such feedback does not imply that a deployed SVM continuously changes its weights in real time. Any proposed feedback dataset must be curated, its demographic/language distribution analyzed, and model updates separately trained, validated and versioned before approval. Neither improved live accuracy nor reduced administrative effort is claimed here without an evaluated deployment study. Residual bias and model-generalization gaps are discussed under Future Work in Chapter 5.

3.3 Research Design and Research Questions

The study uses applied, quantitative experimental design for its completed machine-learning work and system-design methodology for the proposed application. The primary output is an auditable three-class text classifier. The dependent variables are classification accuracy, precision, recall, Macro-F1 and subgroup errors; independent experimental choices include input language assignment, category family, textual length, message construction pattern, TF-IDF feature ranges and SVM regularization. No causal effect of human corrections is

claimed until a frozen pre-update/post-update evaluation is performed.

3.4 Corpus Construction and Provenance

3.4.1 Fixed Research Taxonomy and Label Boundaries

The 118 categories are a scenario inventory and audit framework, not 118 machine-learning output classes. A Valid record conveys an understandable community-related concern or safety report, even if the barangay must refer it to another agency. Suspicious indicates essential context is missing, self-contradictory, or unrecoverably ambiguous *from the input text*; it is not an accusation of dishonesty. Spam communicates no interpretable barangay-reporting purpose. Missing photographs, the absence of a name, nonstandard grammar, or anger alone do not warrant a Suspicious label. Reports *about* scam incidents differ from scam solicitations [1, 4, 11].

Table 4: Three-class operational decision rubric

Class	**Positive criterion and crucial exception**
Valid	Interpretable event, request or community concern. Short, emotional, anonymous and code-switched complaints may all be Valid.
Suspicious	Report-like wording but no recoverable issue, unresolved contradiction, or essential missing reference. It is a *clarification state*, not a finding of falsity.
Spam	Promotional, impersonation or irrelevant submissions lacking citizen reporting intent; also nonsensical or classifier-directed text. A citizen reporting spam or scam activity can be Valid.

3.4.2 Source-to-Category Traceability

The evidence matrix bundled with the Overleaf project maps all 118 category IDs to laws, municipal complaint procedures, literature or researcher-defined rules. The mapping is deliberately qualified: statutes justify topics and jurisdictional context; linguistic papers justify forms of expression; and operational definitions justify unreported or undecidable content. An entry is not considered evidence that a specific synthetically generated narrative occurred in reality. Specialized domains involving violence, children, fraud, utilities and public danger must use appropriate referral practices. An appendix lists the category register used in generation.

3.4.3 Generation Controls and Dataset Composition

The released study corpus was produced with fixed random seed `20261026`. It contains exactly 24,000 unique text records: 8,000 per classification, and within each class 2,000 per generator-assigned language group (Bisaya, Filipino, English, Mixed). The generator uses researcher-authored source-aware scenario descriptions, reason-specific ambiguous

seed messages and non-complaint examples, combined with structured variations in question forms, greetings, follow-up expressions, requested actions, partial location/time cues, emotional language and length bands. Every row is identified as synthetic and unreviewed; its assigned language is not independently verified. The equal distributions are experimentally controlled and cannot estimate true complaint prevalence.

Table 5: Corpus structure and recorded partitions

Dimension	**Count**	**Role**
Valid / Suspicious / Spam	8,000 each	Target labels
Bisaya / Filipino / English / Mixed	6,000 each	Generator allocation
Valid topics / Suspicious reasons / Spam types	60 / 28 / 30	Scenario metadata
Training	19,213	Fit features and select candidates
Validation	2,399	Model selection diagnostics
Synthetic test	2,388	Final within-generator estimate

3.4.4 Data Dictionary and Feature-Exclusion Boundary

The predictive feature is `complaint_text`; the supervised target is `classification`. Fields such as `category_id`, `source_reference_ids`, `scenario_family_id`, `template_family_id`, `language_group`, `split`, and `label_rule` exist only for quality review and experimental controls. Feeding them to the SVM would directly expose the desired label or generation family and invalidate the intended text-only evaluation.

3.5 Data Quality Assurance and Threats to Validity

The generator removed exact-duplicate candidate outputs and verified nonempty narratives, target-class counts and category coverage. These technical checks do not establish natural idiom, correct meaning, grammatical coherence or independent annotation agreement. Accordingly the dataset retains flags marking `language_verified=No` and `human_label_reviewed=No`. The remaining required manual assessment is an adjudicated stratified sample across 118 categories, three classes, four assigned languages, length bands and borderline pairs. At least two independent reviewers should code labels while blinded to generation metadata, adjudicate disagreements, and report agreement alongside confusion rates; a fluent Cebuano/Filipino review is essential before deployment claims. These are planned review tasks, not completed results.

3.6 Experimental Setup and Models

3.6.1 Text Representation and Linear Models

TF-IDF transforms the complaint narratives into a sparse feature space. Multinomial Naive Bayes serves as the baseline estimator using word unigrams/bigrams; the primary model is `LinearSVC` with regularization parameter C. Both operate solely on complaint narrative text, but their recorded TF–IDF settings are different: the Naive Bayes baseline uses word unigrams/bigrams whereas the chosen SVM uses word unigrams. Thus the existing comparison contrasts *complete pipelines* rather than isolating estimator choice with a strictly identical feature matrix. A matched-representation sensitivity comparison can be conducted on the same training/validation partitions in the next documented run; its results must not be attributed to the previous experiment. A `Pipeline` fits the vectorizer within each training fold to avoid using validation or test vocabulary in training estimators [18]. The recorded selected SVM uses word unigrams; an isolated exploratory character 3–5-gram model was evaluated later as an ablation. The combined word-unigram/bigram plus character-within-word 3–5-gram representation is a separately evaluated development candidate, not the feature set producing the historical reference SVM results. Similarly, the reported baseline does not supply fitted calibrated class probabilities. A separate combined-feature sigmoid-calibrated candidate was subsequently evaluated on existing development validation only, as reported in Chapter 4. Independent probability reliability evidence remains required before operational probability-based decisions.

3.6.2 Partitioning and Cross-Validation

The study dataset used scenario-family partitioning before feature fitting: 19,213 train, 2,399 validation, 2,388 test. The 18-config SVM grid combined $C \in \{0.1, 1, 10\}$, TF-IDF word n-gram range in $\{(1, 1), (1, 2)\}$ and minimum document frequency in $\{2, 5, 10\}$. Selection used three-fold `StratifiedGroupKFold` by scenario-family ID with mean Macro-F1. This avoids direct reuse of that ID across folds but does not eliminate shared seed sentences across different family IDs. The best recorded configuration was $C = 0.1$, unigrams, `min_df=2`, with `max_df=0.95` and sublinear term frequency fixed. The observed tie among all 18 grid configurations does *not* demonstrate that tuning has no effect in genuine reporting language.

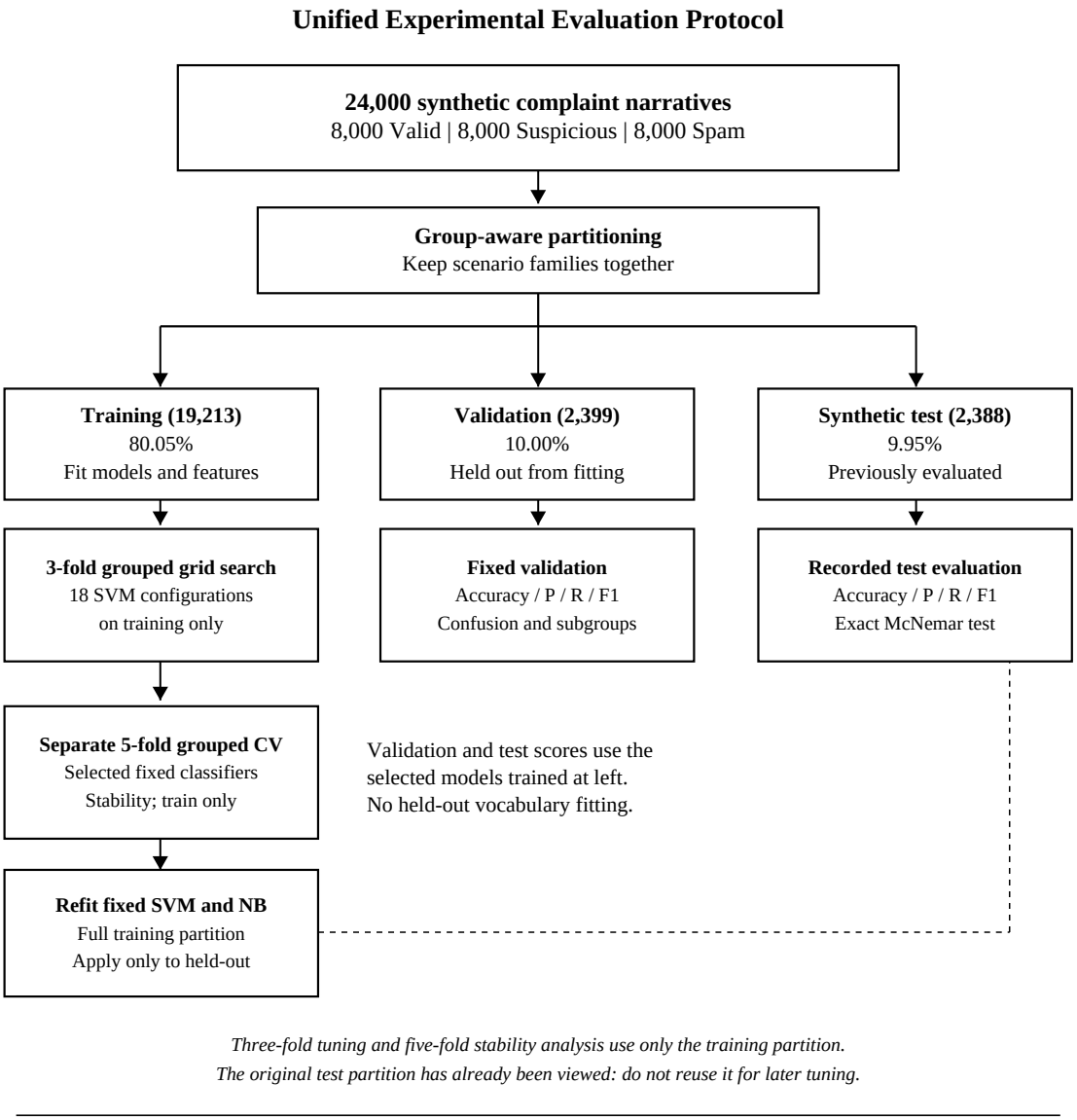


Figure 4: Unified experiment allocation and evaluation protocol. Only the 19,213 training examples participate in model selection and five-fold grouped stability checks; the fixed fitted models are evaluated on 2,399 validation and the previously recorded 2,388 synthetic test examples.

3.6.3 Evaluation and Learning Diagnostics

Accuracy reports the fraction of correct labels, while per-class precision and recall expose specific confusions; Macro-F1 gives equal weight to three classes. Further diagnostics include confusion matrices, class and language subgroup metrics, text-length-only performance, nearest-training-example similarity, and historical challenge summaries. The original challenge input is unavailable for independent re-execution; no replacement challenge

is introduced. Learning curves depict accuracy or Macro-F1 as a function of training-set size; these are *not* loss curves over epochs. The final `LinearSVC` fit does not expose a documented epoch-loss history. Consequently, a separate `SGDClassifier(loss=hinge)` run recorded epoch-wise training and validation hinge loss. This auxiliary trace supports an optimization discussion but must not be mislabeled as the selected SVM model's own training trace.

3.7 Experimental Setup, Reproducibility and Evaluation

The recorded experiment used scenario-family-grouped partitioning (19,213 training, 2,399 validation, 2,388 test), three-fold grouped SVM selection over 18 configurations, and a separate five-fold grouped stability analysis. The following sections specify the completed procedures and distinguish them from prospective operational validation.

3.7.1 Hardware and Software Environment

Table 6 distinguishes the documented original research environment from the environment used for subsequent reproducibility verification. Operating-system patches and library versions may differ between executions; both recorded environments are retained for numerical and procedural repeatability.

Table 6: Recorded research environment and later verification environment

Component	Supplied original record	Later verification record
CPU	Intel Core i7-13700HX	Same CPU model, device query
GPU	NVIDIA RTX 4060 Laptop; Intel UHD	Same GPU models, device query
RAM	16 GB; two 8 GB Samsung DDR5-5600	Two 8 GB Samsung modules at 5600, device query
OS	Windows 11 Home 25H2, build 26200.9457	Windows 11 Home 25H2, build 26200.9711
Python	3.13.5	3.13.5
scikit-learn	1.8.0	1.8.0
NumPy / SciPy	2.3.5 / 1.17.0	2.4.4 / 1.17.1
pandas / joblib	2.2.3 / 1.5.3	3.0.2 / 1.5.3
Plot/QA dependencies	Not fully captured in original record	Matplotlib 3.11.2; Pillow 12.3.0; threadpoolctl 3.6.0

The actual scikit-learn TF–IDF, LinearSVC and Naive Bayes implementations are CPU-based. No CUDA/GPU estimator or GPU acceleration was used by these scripts; presence of an NVIDIA GPU is an inventory fact. Later verification establishes numerical repeatability of recorded outputs, not bitwise reconstruction of the original

environment. The pinned research dependency file represents the verified environment; the supplied historical versions remain identifiable in archival records. Next.js/TypeScript, FastAPI, PostgreSQL and their planned migration/authentication dependencies belong to the proposed application and were not dependencies of the reported offline experiments. pdfLaTeX and the included packages compile the manuscript; their compile logs are artifact evidence, not ML training records.

3.7.2 Version Control and Reproducibility

The research repository is `https://github.com/rencze/nlp-barangay-complaint-classification-hitl`. Published branch `research/milestone2-final-audit`, revision `17ff46846af10f9554a14ec635390865afba50a3`, identifies the reproducibility package. It contains pinned dependencies, scripts, configurations, saved predictions and models, figures and manuscript source. The canonical input is `data/Barangay_Complaint_Dataset_24000.csv`, SHA-256 `35414af2dc63e2870928bd1853c557ed301dbdae80416a0bf37be00648d92e28`; stored partitions define the experimental input.

From the repository root, create an isolated Python 3.13 environment, install `requirements.txt`, and run `pythonsrc/minimal_example.py`. This demonstration prints narrative-to-label predictions; it is not a study-performance estimate. `pythonsrc/verify_dataset.py` checks corpus integrity. Saved evidence is checked by `pythonsrc/final_evidence_audit.py` and `pythonsrc/validate_research_artifacts.py`. The original full-study procedures are `manuscript/source/reproducibility_scripts/evaluate_models.py` and `manuscript/source/validation_suite/run_grouped_validation.py`; exact commands are documented in the repository README. Original outputs must remain preserved when reproducing procedures that write results.

3.7.3 Reproducibility Verification and Execution History

The original archived procedures were executed again during the earlier documentation work; logs, environment manifests, historical output copies and recorded-precision comparisons preserve that history. Those executions supported reproducibility, not a new independent study. Historical outputs, subsequent verification outputs and candidate development outputs are separate named artifacts. No dataset was regenerated, relabeled or repartitioned.

Reference model bundles `results/verified_20261026/svm_research_only.joblib` and `results/verified_20261026/nb_research_only.joblib` contain the matching vectorizer/classifier. `src/evaluate_research.py` reloads a trusted bundle for saved-reference evaluation; reference inference passes raw text to that bundle rather than re-fitting vocabulary or applying the demo helper. These research artifact-loading checks are implemented and verified; their integration into authenticated application submission remains planned. Arbitrary uploaded joblib/pickle files must not be loaded. Prediction exports identify records, partitions, labels and model names; immutable artifact SHA-256 manifests and configurations provide lineage. Final figures are regenerated from actual metrics/predictions as 360-DPI PNG and vector PDF/SVG and presented with academic captions/tables, while CSV/JSON remain audit formats.

3.7.4 Baseline and Measured Model

The standard baseline is TF–IDF plus `MultinomialNB(alpha=1.0)`, a conventional, relatively inexpensive model for sparse text features. The comparison model is word-level TF–IDF plus a linear support vector classifier, `LinearSVC`. Both are evaluated with identical training, validation and synthetic-test partitions; their target labels are Valid, Suspicious and Spam. The historical scores compare two predeclared text-processing pipelines with the same partition memberships, not estimators operating on exactly equal word features. A fair estimator-only follow-up can use the *same* word n-gram representation in both models, fitted on the existing training partition and assessed through training-group folds and the existing development validation partition. Such a follow-up is not part of the recorded results and must be labeled independently. The separately evaluated concatenated word-plus-character representation is *not* the model whose reported test scores appear in Chapter 4.

3.7.5 Evaluation Metrics and Error Analysis

Let $K = 3$ classes, N evaluated narratives, y_i the reference text label of record i, and $\hat{y}_i$ its prediction. For class c, TP_c counts correctly predicted class-c records, FP_c counts other-class records predicted as c, FN_c counts class-c records predicted otherwise, and TN_c counts records neither reference nor predicted as c. Let n_c be reference support in class c and $\mathbf{1}[A]$ equal one when condition A is true, otherwise zero. For single-label multiclass classification:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[y_i = \hat{y}_i] = \frac{\sum_{c=1}^{K} TP_c}{N}, \tag{3.1}$$

$$P_c = \frac{TP_c}{TP_c + FP_c}, \qquad R_c = \frac{TP_c}{TP_c + FN_c}, \tag{3.2}$$

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c} = \frac{2TP_c}{2TP_c + FP_c + FN_c}, \tag{3.3}$$

$$P_{\text{macro}} = \frac{1}{K}\sum_{c} P_c, \qquad R_{\text{macro}} = \frac{1}{K}\sum_{c} R_c, \tag{3.4}$$

$$F_{1,\text{macro}} = \frac{1}{K}\sum_{c} F_{1,c}, \qquad F_{1,\text{weighted}} = \frac{1}{N}\sum_{c} n_c F_{1,c}. \tag{3.5}$$

Zero-denominator class scores are set to zero explicitly. Accuracy expresses overall agreement; precision describes how often predictions of a class are correct; recall describes how many reference examples of that class are recovered. F1 balances precision and recall. Macro-F1 is the primary measure because it gives each of the three operational text labels equal influence rather than letting a common class dominate. Weighted-F1 reflects reference support. Macro-F1 is the mean of class F1 scores, not the F1 of macro-precision and macro-recall.

The confusion entry $M_{ab} = \sum_i \mathbf{1}[y_i = a \wedge \hat{y}_i = b]$ counts reference class a predicted as class b; rows are reference labels and columns are predictions. Per-class precision/recall/F1/support, mistakes, language-group support and confusion counts accompany

aggregate metrics in Chapter 4. A Valid narrative predicted as Spam is particularly important because inappropriate rejection or deprioritization could prevent access to assistance. The proposed routing therefore sends Spam to human review rather than rejecting it automatically. Label agreement neither verifies an incident's factual truth nor measures complaint-resolution effectiveness.

Training-size learning curves measure Macro-F1 as sample count changes. `LinearSVC` does not emit the reported epoch-loss history; the fifteen-pass SGD hinge-loss trace is a distinct auxiliary estimator and must remain labeled accordingly. Length-only and character-only ablations are diagnostic checks for generator shortcuts rather than deployed alternatives. The reported word/character candidate is separately described below.

3.7.6 Actual Hyperparameter Search and Selection

The recorded SVM search used `GridSearchCV` with three-fold, scenario-family-grouped cross-validation on the 19,213 training records. It examined 18 combinations (three C values by two word-ngram choices by three minimum-document-frequency values); all learned TF–IDF statistics were fitted within training folds. The later, separate five-fold `StratifiedGroupKFold` experiment retested the selected settings for stability; it did not perform the original search again.

Table 7: Recorded fixed and varied hyperparameter settings

Setting	Recorded value(s)
`LinearSVC.C` (variable)	$\{0.1, 1, 10\}$
`TfidfVectorizer.ngram_range` (variable)	$\{(1,1),(1,2)\}$ (word analyzer)
`TfidfVectorizer.min_df` (variable)	$\{2, 5, 10\}$
`max_df` / `sublinear_tf` (fixed)	0.95 / enabled
Primary tuning metric	Grouped three-fold mean Macro-F1
Selection from tied candidate scores	$C = 0.1$, word unigrams, `min_df`=2 (deterministic tie handling)
Follow-up stability analysis	Separate five-fold grouped CV on training partition
Random seed	`20261026`

The grouped grid is selected by mean Macro-F1 with deterministic tie handling; observed scores and their interpretation are reported in Chapter 4. The reserved evaluation records were not used to fit TF–IDF or select this grid. The unified allocation and validation protocol is summarized in Figure 4.

3.7.7 Statistical Model Comparison

Because the SVM and Naive Bayes predict the *same* held-out narratives, their correct-versus-incorrect outcomes form paired binary observations. McNemar's test is therefore

appropriate for comparing error occurrence between *two* classifiers on a common test partition [45]. Let b be the count correctly classified by the baseline but wrongly by SVM, and c the count wrongly classified by the baseline but correctly by SVM. The continuity-corrected large-sample statistic and the small-discordance exact alternative are:

$$\chi^2_{\mathrm{McN}} = \frac{\max(0, |b-c|-1)^2}{b+c}, \quad b+c>0, \tag{3.6}$$

$$X \sim \mathrm{Binomial}(b+c, \tfrac{1}{2}), \quad p_{\mathrm{exact}} = \min\Big\{1, 2\Pr\Big[X \le \min(b,c)\Big]\Big\}. \tag{3.7}$$

The hypothesis H_0 is equal paired error probabilities, tested at $\alpha = 0.05$. The exact binomial form is preferred for the present experiment's small number of discordant pairs. Actual paired predictions, contingency counts and interpretations are reported in Chapter 4. The repeated historical test is not a new independent test. A Friedman test would concern a different design involving more classifiers and repeated comparable datasets; it is not justified by the current two-model, one-test-set comparison.

3.7.8 Model Evaluation Sequence

The actual reproducible sequence is (1) preserve the 24,000-row synthetic source, codebook and scenario-group IDs; (2) group-partition into 19,213/2,399/2,388; (3) fit the baseline and perform three-fold grouped SVM grid search on training only; (4) conduct a separate five-fold grouped stability analysis on the training portion; (5) evaluate the selected fixed models on 2,399 validation records and the previously evaluated 2,388 synthetic test records; (6) report class-wise and aggregate scores, confusion matrices and exact paired-error comparison; and (7) document scripts, seed, versions, selected parameters, generator limitations and the limitation that performance outside this synthetic corpus is not measured. Chapter 4 contains *observed results*; combined features and calibration are separately evaluated development candidates, while operational HITL and model retraining remain proposed system capabilities.

3.8 Reproducibility Protocol and Evidence Boundary

For the completed rerun, the researchers preserved the supplied `ALL_SYNTHETIC_V2.csv` and its recorded partition memberships. **The current repeatability exercise uses the same 24,000-record synthetic corpus; no additional dataset or relabeling is part of this reproduction.** The set comprises 19,213 rows for fitting and grouped cross-validation, 2,399 rows for fixed development validation, and 2,388 rows for the historically evaluated synthetic test. Only the training partition is used for estimator and TF–IDF fitting. A validation score is not a training score, and a repeated score from the same test rows is not new external evidence.

The GitHub artifact should record a Git commit identifier, the SHA-256 of the 24,000-row source file, Python and library versions, experiment seeds and exact commands. The dataset partitions are exported from the `split` column without generating or resampling records, and a preflight script checks disjoint scenario-family IDs and counts before model fitting. The two principal reproducible commands are

`pythonreproducibility_scripts/evaluate_models.py` for the original model comparison and `pythonvalidation_suite/run_grouped_validation.py` for the separate five-fold training-only stability audit. The repository also includes a small smoke test, which is not evidence of full-study accuracy. Since the full scripts may overwrite recorded outputs, the original result files must be archived and versioned before a rerun. Any discrepancy is logged, not silently rewritten into the thesis.

Reproducibility verification established whether the recorded preprocessing, SVM, Naive Bayes and cross-validation scripts execute on unchanged splits. If a second, explicitly labeled *sensitivity experiment* compares a common TF–IDF representation or the proposed word–character combination, it must use training-group folds and the existing development validation partition for configuration decisions. It must not use the previously viewed synthetic test as an unseen selection set. Report the new settings and results separately instead of replacing the historical scores. A subsequent combined-feature/calibration experiment was executed on development validation only; operational HITL, feedback retraining and live case management remain untested regardless of their appearance in design diagrams.

3.8.1 Reproducibility Artifacts

Table 8: Reproducibility artifacts and what each establishes

Artifact	Required evidence
Source CSV and split files	Source hash; partition totals; label counts; no scenario-family overlap across partitions.
Baseline and SVM scripts	Refit TF–IDF within each training fold; record selected parameters and seeds.
Validation metrics	Accuracy, class-wise precision, recall, F1, Macro-F1, weighted-F1 and confusion matrices on 2,399 records.
Synthetic test archive	Historical scores on 2,388 records, explicitly marked as already evaluated; preserve paired predictions if available.
Five-fold stability audit	Training-only grouped folds; group overlap report and fold-wise metrics.
README, environment and commit	Installation instructions, exact commands, dependency versions and source-code revision.

3.9 Required Next Validation Protocol

As *future validation recommendations*, rather than conditions claimed to have been completed in the current single-corpus rerun, robust deployment evaluation would include: (a) independent blinded human label agreement; (b) broader unseen-scenario / root-seed holdout independent of template families; (c) deliberately constructed minimal-pair tests of ambiguous versus clear narratives; (d) subgroup performance on Bisaya, Filipino, English and mixed inputs; (e) safety-sensitive false-negative review; (f) probability/margin

reliability and an abstention or "review required" policy; and (g) full-stack end-to-end tests for accounts, audit logs, urgency escalation, status transitions, feedback consent, versioning, latency and rollback. These items are not evidence generated by the current experiment. The immediate documented rerun uses only the existing synthetic corpus and its stored train/validation/test membership; any subsequent external assessment, if undertaken, must be reported as a separate study.

3.10 System Architecture and Operational Boundary

The proposed system distinguishes a researcher-evaluated word-level TF–IDF/SVM baseline from a planned operational architecture that adds privacy-aware input preparation, class-based HITL review, accepted-case management and controlled retraining. Word-plus-character features and sigmoid calibration were separately evaluated on synthetic development validation; they do not establish operational reliability. The measured model scores are not evidence that the additional production functions are already implemented or validated. Section 3.10 defines these interfaces, states and non-automatic feedback processes. System-level monitoring and administrator accountability are application requirements to be evaluated separately from classifier accuracy.

Implementation boundary. Figures in this section specify the *target application design*. They are not additional completed experiments. The reproducible results in this manuscript concern the existing synthetic-data word-level TF–IDF and Linear SVM pipeline. The combined word/character representation and grouped sigmoid calibration were separately tested on existing synthetic development validation. Live probability reliability, class-dependent administrative review, case management and feedback-driven retraining still require independent implementation and measurement before production-readiness claims. Each diagram is placed immediately after the explanatory discussion of its named process; figure numbers are assigned by LaTeX captions rather than embedded in the graphics.

3.10.1 Complaint Intake and Input Boundary

The proposed resident-side form requires a *Subject* and a *Complaint Narrative*. If known, a date/time of occurrence and an approximate Purok/Sitio or landmark can be supplied as supporting context, without compulsory GPS or an interactive map. The narrative, not the user's name, phone number, assigned language tag or other metadata, is the primary input to the three-class SVM. The subject can be retained for case identification, but adding it to the classifier requires a separately documented experiment to preserve comparison with the recorded text-only baseline. The application should provide a confirmation and tracking reference and maintain authenticated access and lawful handling of resident information. The screenshots of Davao City Reports serve as a *local interface precedent*, not evidence that its implementation has been replicated [7, 32].

The core scope excludes photo uploads, image classification, ensembles that combine images and text, automated determination of physical truth and mandatory GPS coordinates. Where urgent safety reports arise, an independent referral/escalation safeguard

is recommended rather than allowing an ordinary three-class text prediction to prevent timely review. Such referral logic remains a system requirement, not an experimentally demonstrated function.

3.10.2 NLP Preprocessing: Meaning-Preserving Candidate

Figure 5 specifies a proposed conservative NLP routine. A secure original can be retained for authorized processing while the model operates on a minimized copy with direct identifiers masked when appropriate. The depicted privacy-oriented operational routine uses Unicode NFC, whitespace normalization, lowercasing and limited cleanup; these are intended to standardize representation without discarding *hindi*, *wala*, *no*, *not*, relevant number expressions or incident descriptions. Generic stopword removal, stemming and lemmatization are excluded from this proposed routine to avoid meaning-altering changes to mixed-language complaints. The current reproducibility scripts use scikit-learn TF–IDF's text processing; the separately depicted privacy masking and normalization stages are not asserted to be fully implemented in those recorded experiments. The separately evaluated development candidate instead uses NFKC/casefold/whitespace and no identifier masking, as specified in Section 3.11.1; these preprocessing variants must not be interchanged silently. Any masking accuracy must be assessed because over-masking can remove useful context and under-masking can expose personal information [32, 23].

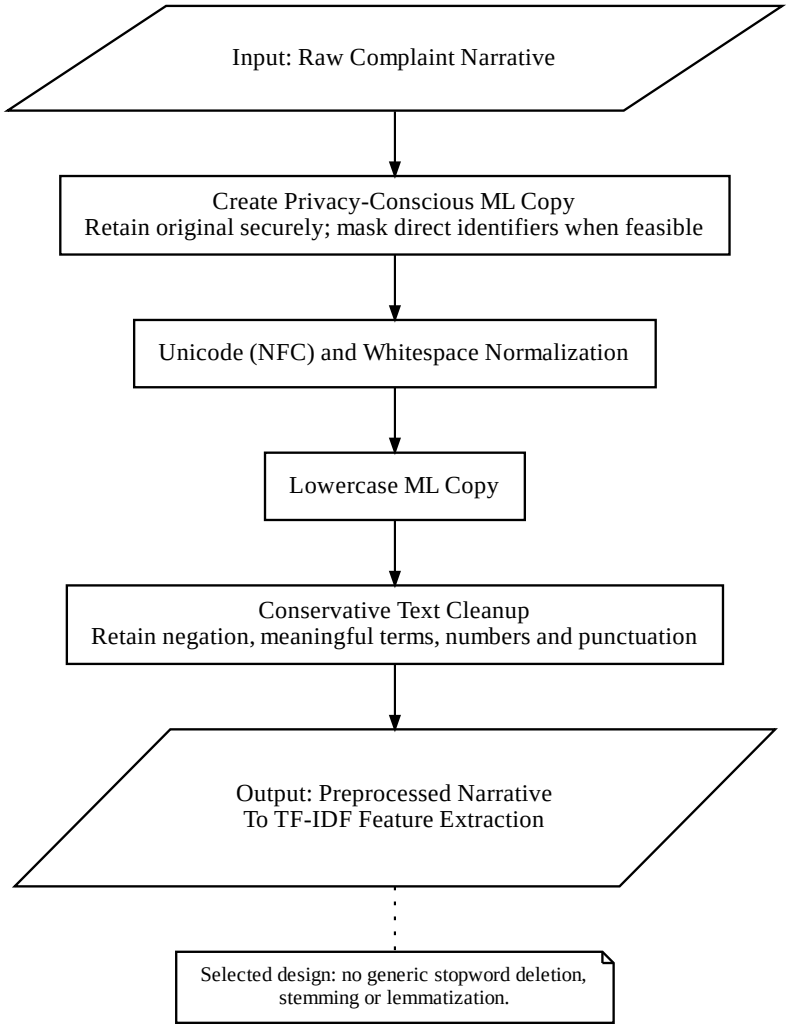


Figure 5: Proposed Natural Language Processing (NLP) preprocessing workflow for complaint narratives. Privacy-aware masking and normalization are target features, not measured preprocessing steps in the reported synthetic-data experiments.

3.10.3 TF–IDF Feature Extraction and Model-Level Consistency

The recorded main SVM results used word-unigram TF–IDF, with a separately examined character-only ablation. Figure 6 describes the separately evaluated candidate using word unigrams and bigrams *together with* character-within-word 3–5-grams. The resulting two sparse feature matrices are concatenated before classification. This combination is a *separately evaluated development candidate*, not the verified configuration yielding the previously reported perfect within-generator score. In both configurations, vocabulary and inverse-document-frequency weights must be fitted within training partitions, never

independently refitted on each new complaint. The deployed vectorizer and classifier should be versioned together [43, 23].

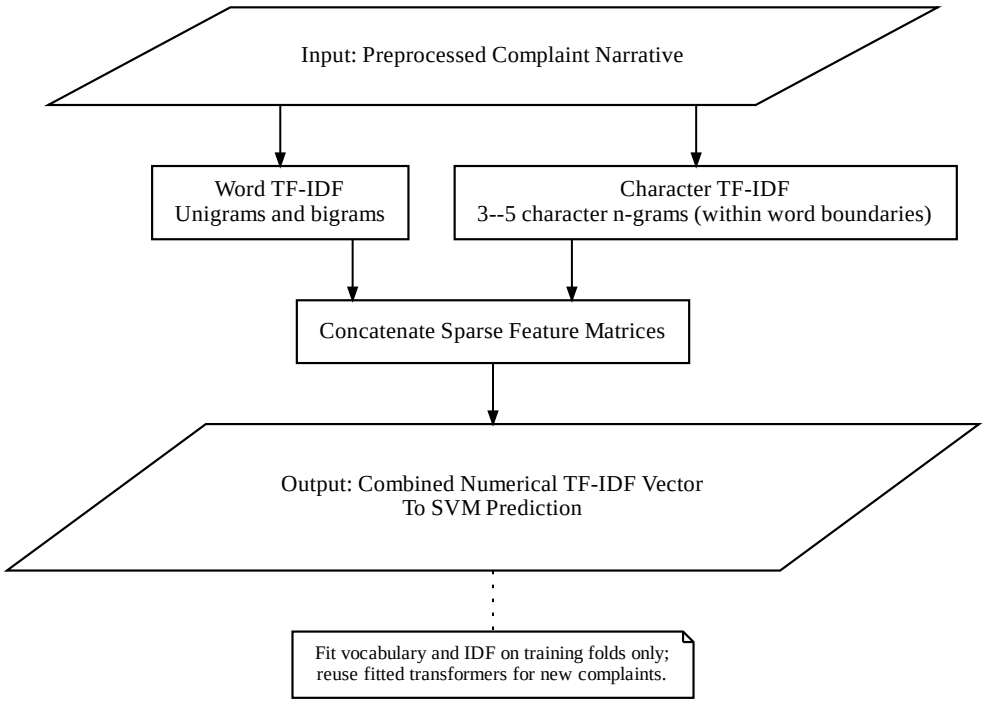


Figure 6: Separately evaluated word–character TF–IDF candidate. Concatenation and grouped sigmoid calibration were evaluated on development validation and tied the original word-only SVM Macro-F1; operational deployment and external reliability remain unvalidated.

3.10.4 SVM Training and Calibrated Prediction

Figure 7 separates model fitting from inference. An offline Linear SVM is trained using authorized labeled data, with hyperparameter selection performed through training-only cross-validation. Before model probabilities can be described as *calibrated*, an appropriate calibrator must be fitted using independent calibration folds or held-out training material, then evaluated for reliability. During inference, a new complaint is transformed by the saved compatible vectorizer and classified by the saved model; it is not used to fit the model. The three displayed class-probability quantities remain proposed operational outputs; the reference SVM is uncalibrated, while a separate candidate supplies synthetic development probability diagnostics in Chapter 4 [44, 23].

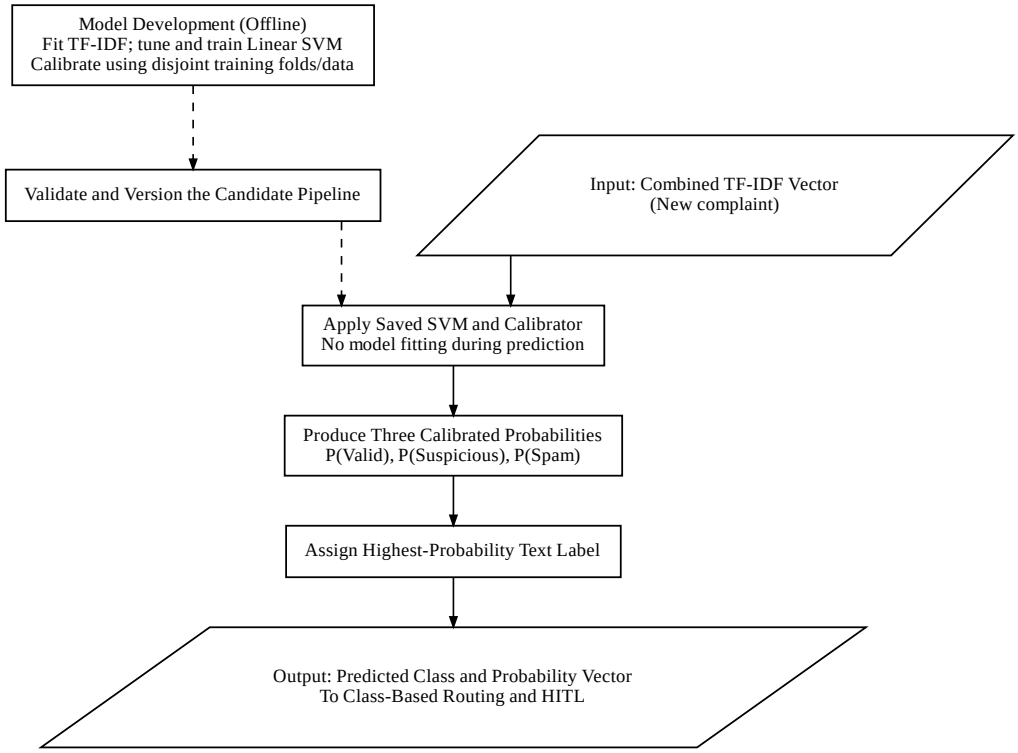


Figure 7: Proposed support vector machine (SVM) model-development and new-complaint prediction workflow, with a separate probability-calibration requirement.

3.10.5 Human-in-the-Loop Verification and Administrative Disposition

The initial proposed routing rule is *class-based*: predictions of Valid go directly to accepted-case handling, while predicted Suspicious or Spam are referred to the administrator for Human-in-the-Loop (HITL) review. Calibrated scores, once validated, may help *prioritize* cases within the review queue; no numerical cutoff is fixed by this study. Under this design, an error that predicts Valid will not enter routine HITL, so a separately selected random audit of predicted Valid cases is necessary to estimate unobserved misclassifications. This is a proposed mitigation of the selective-label problem, not an implemented audit [36, 37].

The administrator first examines the particular original text version and may confirm or correct its *text label* to Valid, Suspicious or Spam. Separately, the administrator makes the *disposition*: Accepted, Pending Clarification or Rejected with documented reason. Figure 8 shows the decision boundary. A predicted Spam that was actually a clear complaint can be corrected to Valid and accepted; an incomplete original can remain Suspicious while external clarification permits administrative acceptance. No accepted case is automatically converted to a Valid training label, and no rejected case is automatically converted to Spam. A clarification request must be reassessed after a response or the

applicable policy-governed response period; no invented universal deadline is imposed [33, 34, 38].

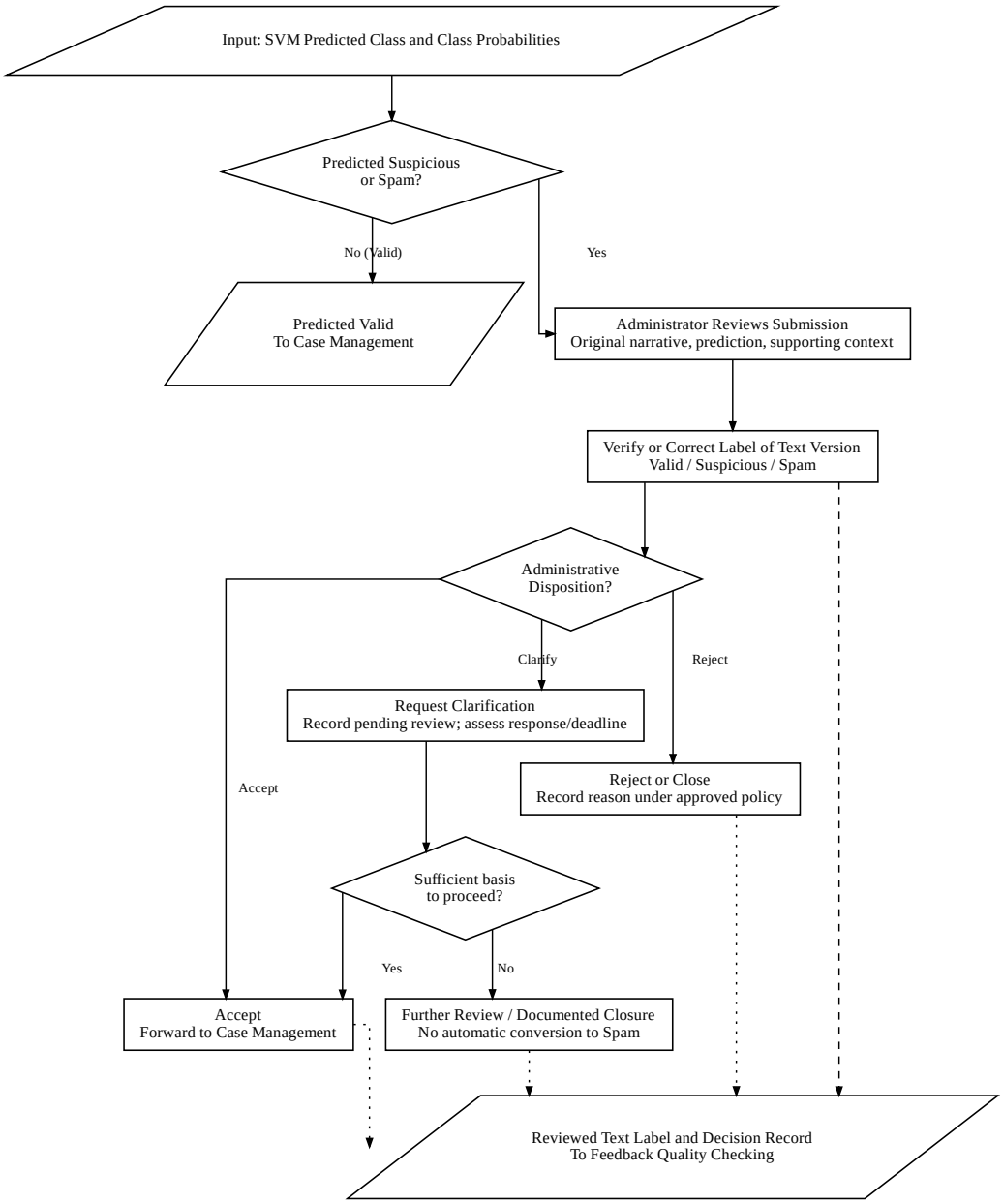


Figure 8: Proposed Human-in-the-Loop complaint review with distinct text-label verification and administrative disposition. Only predicted Suspicious or Spam enter routine HITL; direct Valid predictions proceed separately.

3.10.6 Nine Boundary Scenarios for Consistent Labeling

Table 9 operationalizes the ambiguous cases found in the consolidated evidence review. The text label applies to the *narrative version*, not the independently investigated incident or the administrative case outcome. These are annotation and workflow rules to be validated by reviewers, not performance findings from an authentic dataset [33, 34, 35].

Table 9: Proposed boundary-case labeling and administrative handling rules

Scenario	Verified narrative label	Administrative handling
Predicted Suspicious, but original text is already clear.	Correct to Valid.	Accept and route to case management.
Original is vague; resident supplies complete revision.	Original Suspicious; revised Valid.	Accept revised narrative; preserve both versions.
Original vague; independent evidence supports proceeding.	Original remains Suspicious.	May accept if authorized procedure permits.
Predicted Spam, but original is an actionable complaint.	Correct to Valid.	Accept; do not automatically reject.
Predicted Spam, but original is complaint-related and incomplete.	Correct to Suspicious.	Clarify; accept if sufficient information follows.
Predicted Suspicious, but content is promotional non-complaint material.	Correct to Spam.	Reject after human confirmation and record reason.
Text is clear but allegation has not been substantiated.	Valid as text.	Investigate or document outcome; truth not inferred by SVM.
Clear text later has legitimate administrative grounds for closure or rejection.	Valid as text.	Record grounds without changing text label.
Repeated similar submissions from one or several residents.	Label each version by its text.	Check duplicates and corroboration independently.

3.10.7 Verified Feedback, Data Quality and Controlled Model Retraining

Figure 9 begins with versioned human-reviewed label records; it does not repeat the administrative Accept/Reject workflow. Records first undergo quality, duplicate, provenance, privacy and split-family checks. They are then accumulated in a verified-feedback pool. A periodic review of count, verified-class distribution, diversity, repeated errors and performance degradation determines whether a candidate training run is justified; the run requires authorization. Retraining is *not triggered by each click* on an administrative button. An approved run consolidates the existing eligible training material and the new verified examples, refits the preprocessing and vectorization components, trains and

recalibrates a candidate SVM, and evaluates it on a fixed independent validation protocol. A candidate that fails release criteria is archived without replacing the live model; a candidate that passes can be separately approved and versioned for future predictions [39, 41, 23, 22].

The consolidated research review proposes cumulative increments of *100, 250 and 500* verified examples as learning-curve checkpoints, with 1,000 only if feasible. They are *research planning choices*, not empirically validated minimums or automatic deployment triggers. The baseline plus those three candidates are four conceptual model configurations, but cross-validation and calibration can require many additional fits. All variants of one case and related synthetic templates must remain grouped to prevent evaluation leakage. A small independent audit of predicted Valid cases should prevent the feedback pool from consisting solely of Suspicious/Spam predictions. Existing synthetic-data Macro-F1 scores cannot demonstrate that retraining on authentic complaints improves generalization [39, 40, 42, 41, 36].

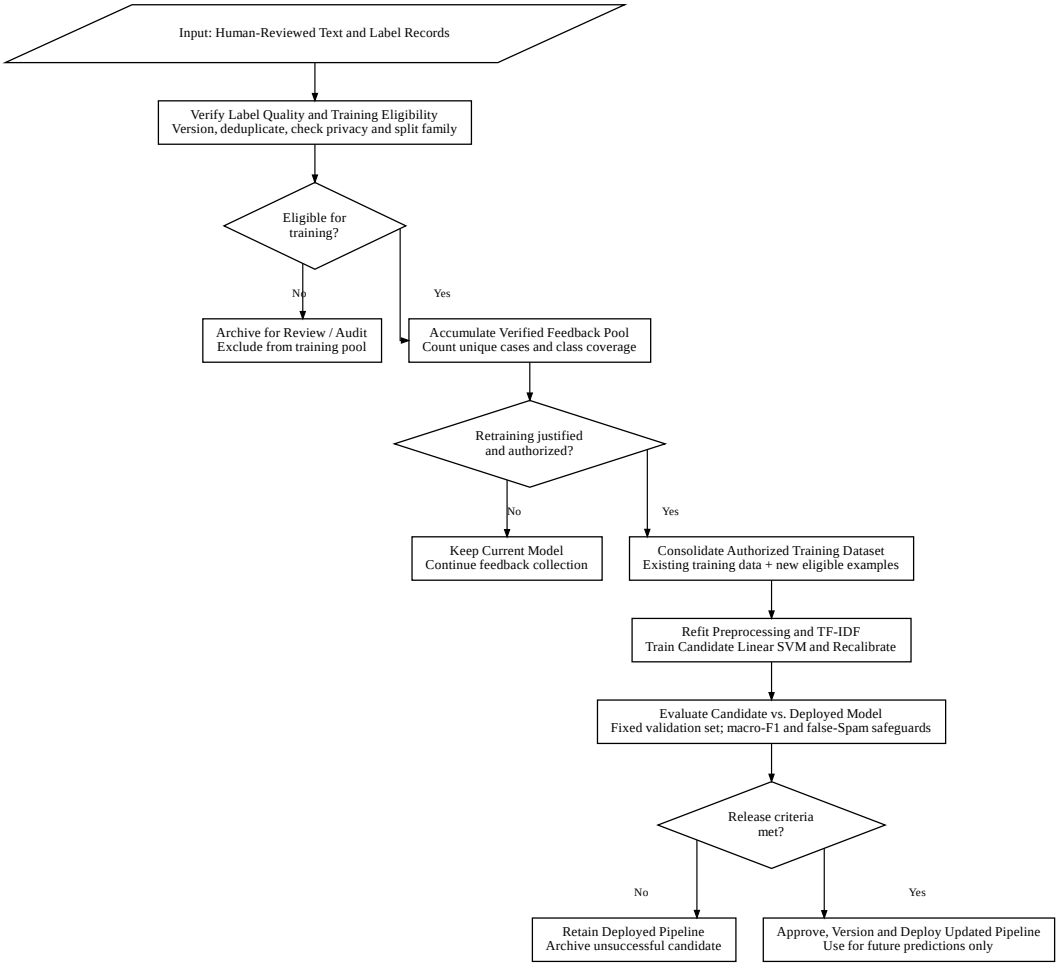


Figure 9: Proposed quality-checked human feedback and controlled batch retraining; training eligibility and deployment approval are separate decisions, and failure does not trigger infinite retries.

3.10.8 Accountable Complaint Case Management (Accepted Cases Only)

The accepted-case workflow in Figure 10 begins after the system accepts a predicted Valid narrative or an authorized administrator accepts a reviewed submission. The case receives a tracking identifier and a principal state of *Unresolved*; designation of responsible staff, documented investigation and progress updates move the case to *Ongoing*; independently checked completion allows it to become *Resolved*. Records capture officer assignment, action timestamps, correspondence and supporting justification. *Overdue*, *Referred* and *Reopened* are auditable events or flags, not additional SVM labels. If an action is unfinished, disputed or inadequately verified, the case returns to follow-up rather than being silently finalized.

This proposed administrative design is grounded in the distinction between public *response obligations* and full incident resolution. Republic Act No. 6713, Section 5(a), addresses replies to public communications within fifteen working days; Republic Act No. 11032 and the relevant Citizen's Charter govern processing of covered services by complexity. The special 72-hour concrete-action standard associated with the national 8888 channel is *not* a universal deadline for resolving every barangay incident. The application must store the locally applicable rule, permit reasoned referral/escalation and support resident disagreement. Davao City Reports provides a local precedent for tracking references, progress history and re-filing, not a claim that this proposed barangay implementation already meets its requirements [29, 30, 31, 7].

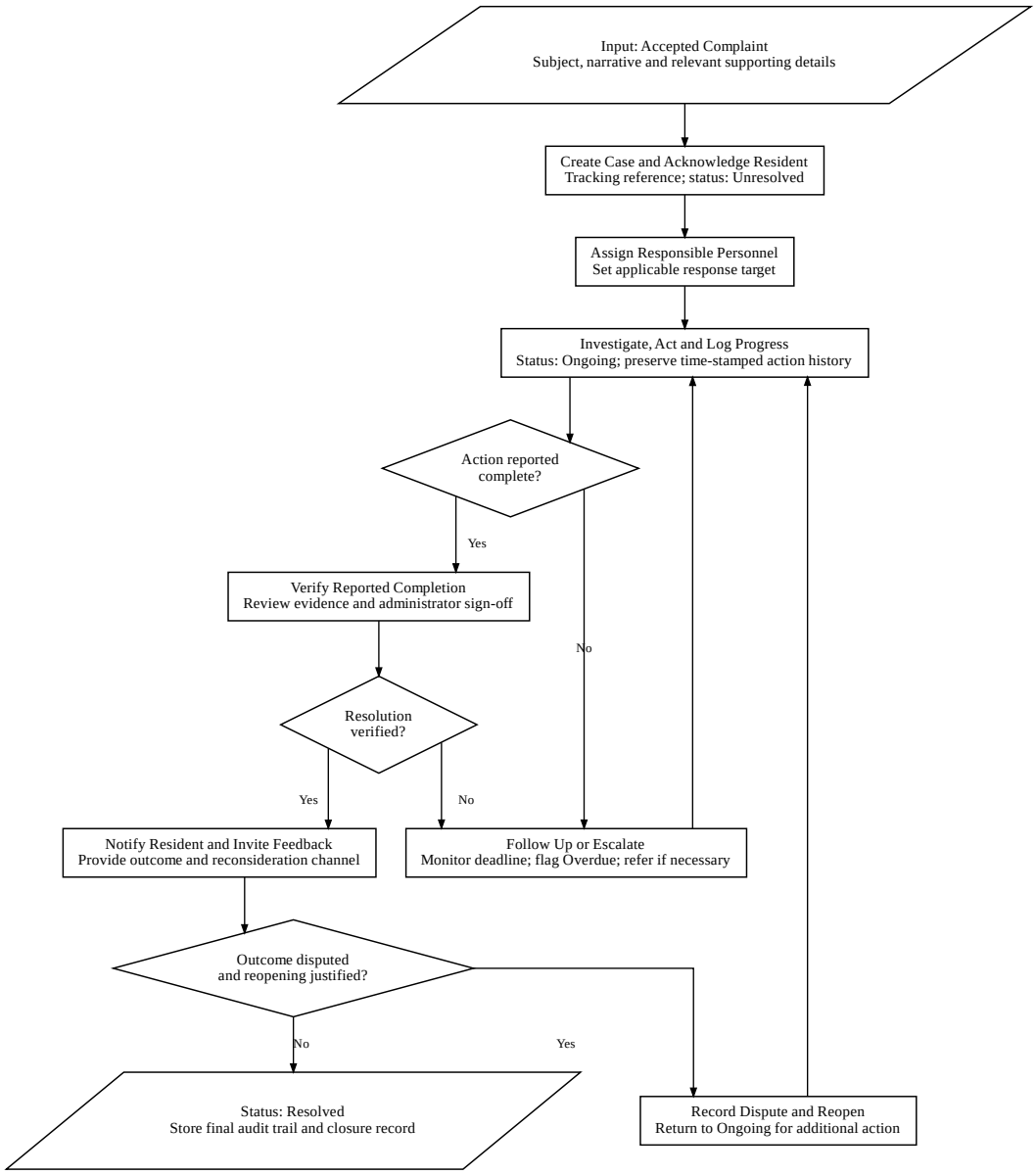


Figure 10: Proposed accountable complaint case management: accepted-case tracking, action logging, policy-based follow-up, completion verification, resident feedback and justified reopening.

3.10.9 End-to-End Integrated Complaint Workflow

The integrated workflow in Figure 11 links the previously described modules into one end-to-end account of proposed system behavior. A complaint narrative passes through conservative NLP preprocessing, saved word/character TF–IDF transformers and a calibrated SVM. Predicted Valid narratives proceed to accepted-case management; only predicted Suspicious or Spam narratives enter routine HITL for text-label verification

and a separate administrative decision. Reviewed but non-accepted submissions retain a documented clarification or rejection outcome. *Accepted* is an administrative state, not a replacement for the narrative label, while *Resolved* is a later case-management state, not proof that an originally Suspicious sentence was Valid.

The diagram intentionally separates live case handling from the offline learning loop. Case records store assignment, action history, response deadlines and resolution outcomes, whereas versioned human-verified narrative labels are quality-checked before they can enter the training-feedback pool. Training a new TF–IDF–SVM candidate requires a justified and authorized retraining decision; the deployed classifier changes only after successful validation and release approval. In particular, neither an Accept click nor case resolution automatically updates model weights or training labels. These are proposed control boundaries, not additional observed results from the recorded synthetic-data baseline [32, 33, 23, 22].

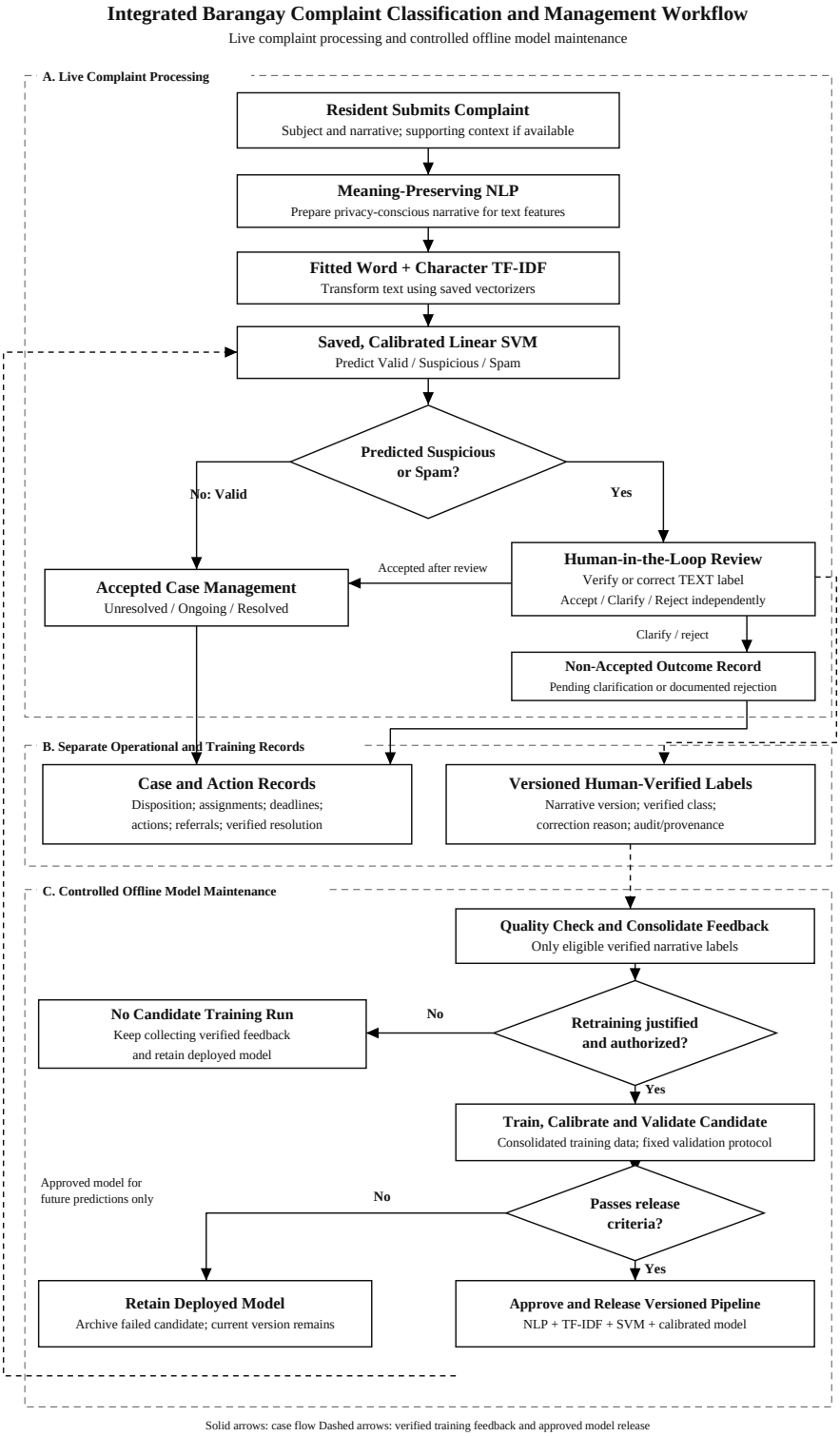


Figure 11: Proposed end-to-end integrated workflow connecting resident submission, NLP, TF–IDF, SVM, class-specific human review, accepted-case processing and approved offline retraining. Dashed arrows distinguish training-feedback and approved model updates from live complaint handling.

3.10.10 Software Component Integration and Data Boundaries

After the end-to-end workflow, Figure 12 presents the complementary *software component integration* and its access boundaries. The proposed resident portal and authorized administrator dashboard exchange data through a FastAPI application layer rather than calling the classifier or writing database records directly. The backend manages access checks, request validation, classification requests, tracking references, administrative actions and auditable case updates. Next.js/TypeScript, FastAPI/Python and PostgreSQL are the suggested application stack; the uploaded reproducibility evidence does not demonstrate deployment of these services.

The text-only inference service consumes the complaint narrative, applies the saved compatible preprocessing and TF–IDF transformations, and returns a saved calibrated SVM's proposed class probabilities. Predicted Valid follows the accepted-case service; predicted Suspicious or Spam enters administrator review. Reviewers may correct the *text label* independently of Accept/Clarify/Reject decisions. Accepted cases and non-accepted submission outcomes share secured persistence but maintain different administrative states, while eligible verified narrative labels are stored as a *distinct feedback dataset*. Later, an authorized offline candidate training run consolidates eligible training records, retrains the feature extractors and classifier, checks calibration and held-out validation, and releases a single versioned model bundle only after approval. The approved release path denotes an approved future inference version, never incremental training during complaint submission. The proposed design excludes photographic inputs and location maps from model inference and does not claim that a completed case is necessarily a Valid textual training example [23, 44, 32, 22].

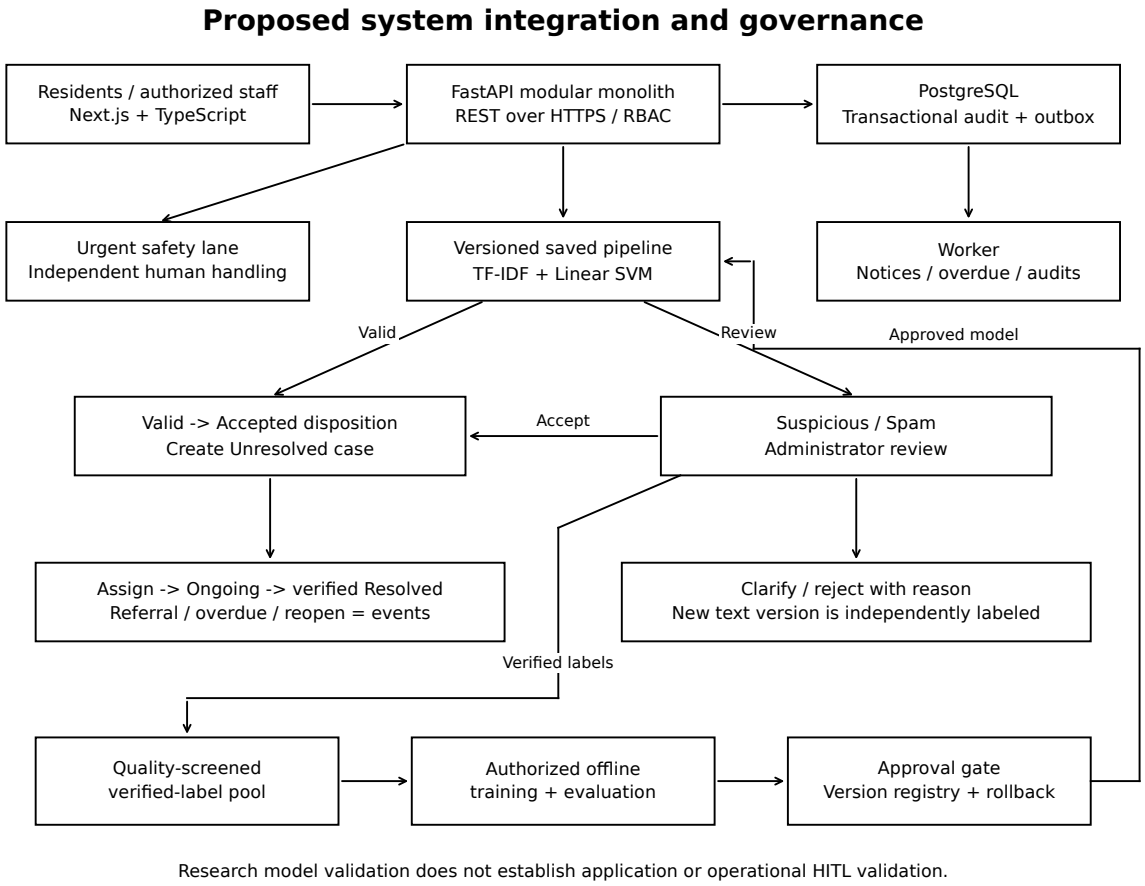


Figure 12: Proposed system integration architecture showing the resident and administrator interfaces, application API, text-classification service, HITL review, separate case and feedback records, and controlled offline model-release boundary. The component diagram specifies a target implementation, not a deployed or evaluated full-stack system.

3.10.11 Implementation and Evaluation Gates

Table 10: Evidence boundary and evaluation required for proposed extensions

Component	What the supplied research currently supports	Required next evidence
24,000-row synthetic dataset	Fixed counts, partitions, metadata and researcher-generated texts; all review flags negative.	Human annotation and authentic external validation.
Word TF–IDF + Linear SVM	Recorded group-aware grid/CV and within-generator validation; test generalization remains limited.	Independent test beyond generator patterns.
Word + character combination	Reference reproduction and separate combined-feature development experiment.	Independent external comparison beyond the synthetic ceiling.
Probability calibration and HITL	Synthetic development calibration; HITL remains a proposed policy.	External probability reliability, reviewer agreement and Valid-sample audits.
Human-feedback retraining	Cumulative checkpoints and release-gate plan.	Verified feedback, fixed validation, approved candidate run.
Case management/accountability	Flowchart and local legal/interface precedents.	Implemented audit logs, deadlines, referral and resident tests.

3.10.12 Detailed Full-Stack Implementation Plan

The proposed architecture separates Next.js/TypeScript resident and administrator interfaces, FastAPI inference and workflow endpoints, and PostgreSQL records for versioned narratives, predictions, verified labels, dispositions and accepted cases. Object-scoped authorization, append-only histories, explicit state transitions, and model-release approval are required application controls. Figure 12 supplies the single software-integration overview; Figure 13 gives the complementary relational model. These are implementation plans, not validation of the existing SQLite/static-HTML prototype. Direct Valid routing records policy-based acceptance but never creates a human-verified label. Urgent human handling is independent of predicted class. The editable diagram sources and detailed schema are maintained in the repository documentation. The relational design links immutable narrative versions to separate predictions and verified labels; accepted dispositions alone create cases. Append-only action and audit histories preserve the distinction between machine outputs and administrative decisions.

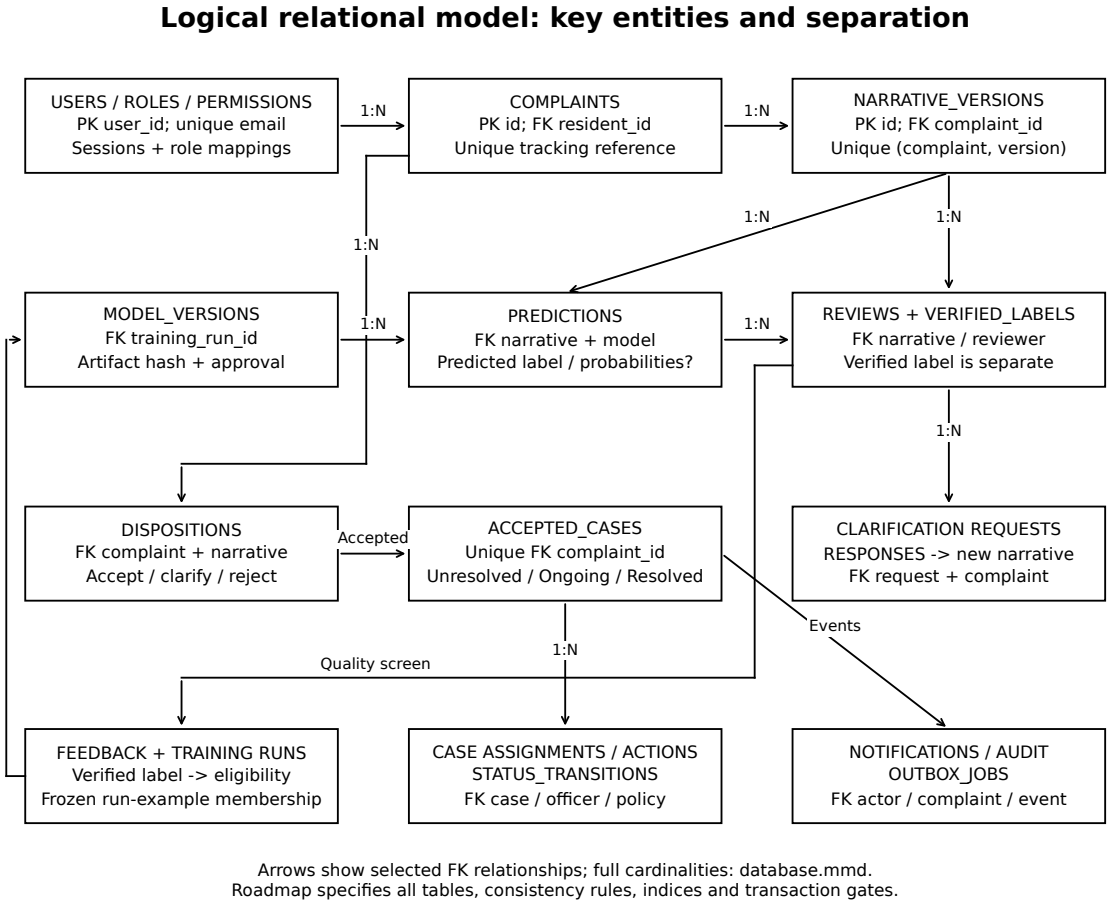


Figure 13: Proposed logical relational overview; keys, constraints, indices and cardinalities are specified in the repository schema design.

3.11 Research Ethics, Reproducibility and Limitations

The corpus contains researcher-generated narratives rather than private resident reports. Reproducible inputs include the category register, generator source, random seed, partition CSVs, run records, training scripts and model metrics. Ethical deployment requires privacy-conscious logging and retention, a clear review process, and transparent handling of urgent reports. Because generator-authored patterns remain correlated with class labels, the measured within-generator accuracy can exaggerate transferable performance. No prospective performance gain from Human-in-the-Loop retraining or reduction in office workload is claimed without an implemented and evaluated user study.

3.11.1 Separately Evaluated Combined-Feature Candidate

The candidate is distinct from the original SVM: Unicode NFKC, casefolding and whitespace normalization preserve negations and accented characters; word $(1, 2)$ TF–IDF

(min_df 2, max_df .95) is concatenated with character-within-word $(3, 5)$ TF–IDF (min_df 3, maximum 45,000 features). Both branches use sublinear TF and default L2 normalization, equal weights and LinearSVC $C = .1$. One fixed candidate was evaluated; no new search was performed. A calibrated variant uses sigmoid `CalibratedClassifierCV` with an ensemble of three precomputed scenario-disjoint training folds, each refitting its complete feature pipeline. Calibration folds do not fit their corresponding base estimators. Existing development validation evaluates the candidates; the historical test was not used for them [44, 23].

Both variants tied the reference's validation Macro-F1, so no classification improvement is demonstrated. Log loss, multiclass Brier score and ten-bin expected calibration error describe synthetic development probability behavior. These are completed development diagnostics, not independently verified operational calibration. The reference SVM has no calibrated probabilities; routing remains class-based, not an untested probability threshold. Chapter 4 reports candidate findings separately; live calibration approval and external validation remain future work.

3.12 Research-to-Experiment Traceability

The study's single two-phase conceptual framework is presented earlier in Chapter 3. For the completed experimental phase, researchers selected scenario categories, generated a controlled corpus, partitioned related scenarios before fitting text features, fitted the fixed baseline and tuned the SVM on the training partition, and evaluated the fixed models on separately held-out partitions. These steps are explained here without repeating the conceptual framework diagram. The single combined evaluation-protocol figure is Figure 4.

3.13 Data Source and Scenario Acquisition Protocol

The project does not claim to have collected 24,000 authentic resident reports. *Data acquisition* refers here to acquiring a documented *evidence foundation* for generating fictional, text-only scenarios. The category register has 60 report-topic entries, 28 clarification/review-reason entries, and 30 spam/non-complaint entries. The source-to-category matrix links each entry to a suitable law, administrative reporting practice, linguistic study, or operational labeling rule. For example, the Local Government Code [1] provides the broad basis for public-service reporting; RA 9003 [2] provides subject-specific environmental context; Davao City Reports [7] illustrates expected descriptions of location and incident details; and the Filipino complaint-expression studies [8, 10] support alternatives to rigid report templates. An authority that establishes the existence of a reporting topic does *not* establish the relative frequency or authenticity of each synthetic narrative.

Table 11: Documented roles of evidence used in controlled generation

Evidence class	Generation decision supported	Limit of inference
Philippine legislation	Legal or administrative relevance of a public concern	Does not define the three model labels
Official complaint workflows	Common incident fields and missing-context issues	Cannot prove a given report is false
Complaint-pragmatics research	Directness, criticism, questions, requests and affect	Does not provide nationwide style proportions
Code-switching research	Plausible multilingual phrase ordering and lexical borrowing	Not direct evidence of barangay frequencies
Scam and spam studies	Promotional requests, impersonation and solicitations	Does not establish all unrelated-message types
Synthetic data research	Diversity control, documentation and independent evaluation	Does not guarantee field generalization

3.14 Labeling Codebook and Error-Sensitive Boundaries

The primary label is operational and text-based: **Valid** indicates an understandable report of a civic issue or a request for action, regardless of whether the barangay itself has jurisdiction. **Suspicious** indicates that a plausible report cannot be interpreted sufficiently because essential meaning is missing or internally contradictory. **Spam** indicates an irrelevant, deceptive, promotional, or non-complaint submission. These labels do not establish truthfulness or determine legal jurisdiction. A brief Cebuano report with a clear problem is not Suspicious because of brevity alone; a complaint about someone attempting a scam is not Spam merely because it contains scam-related vocabulary. The codebook and category-level evidence are supplied with the dataset.

If a condition depends on information the classifier never receives (for example, an absent file attachment or a separate form field), the system should assign the check to application validation or a human reviewer rather than learning an inference unsupported by `complaint_text`. Urgent public-safety and protection-related cases must not be blocked solely by a classifier label. This is a design safeguard, not an evaluated software behavior in the present study.

3.15 Two-Level Validation Design

For the present study, the experiment uses **two distinct grouped resampling procedures**. First, a three-fold `GridSearchCV` searches 18 SVM configurations on the 19,213 training samples. Second, an additional five-fold `StratifiedGroupKFold` is run on the same training partition for the fixed chosen SVM and Naive Bayes baseline to report fold-specific stability. The groups are `scenario_family_id` identifiers, not the three class labels, and each held-out fold has zero overlapping group IDs with its corresponding training fold. The vectorizer is fitted anew inside every fold. The held-out validation set comprises 2,399 examples and is used for validation metrics, class confusion, language subgroup comparisons, and descriptive feature-ablation diagnostics. The 2,388-record final test set was excluded from the supplementary five-fold suite; it had already been evaluated in the original full-results experiment and is not an untouched future test set.

3.16 Evaluation Score Reporting and Fold Dispersion

The primary scoring metric is Macro-F1; Section 3.7.5 defines accuracy, class precision, recall, F1, macro- and weighted-F1, and the confusion matrix. For the separate five-fold stability analysis, the arithmetic fold mean is $\bar{F} = \frac{1}{5}\sum_{i=1}^{5} F_i$ and the sample standard deviation is $s = \sqrt{\sum_i (F_i - \bar{F})^2/4}$. These describe dispersion across generated-data folds; they are not independent population confidence intervals.

3.17 Reproducibility and Validation Integrity

The accompanying grouped-validation script includes assertions for the 24,000-record count, training/validation group separation, and zero within-fold group overlap. It exports the fold composition, five-fold metrics, validation predictions, per-class and language-subgroup metrics, confusion matrices, and diagnostic ablations. The recorded seed is `20261026`, and the environment uses scikit-learn version 1.8.0. The full experiment still depends on the stated generator and label-rubric assumptions. The additional recorded diagnostics must be interpreted as a retrospective validation supplement and not as a preregistered independent test.

Chapter 4

Results and Discussion

4.1 Experimental Scope and Documentation

This chapter presents the original recorded research on the fixed 24,000-narrative synthetic corpus: group-aware SVM selection, a Naive Bayes pipeline baseline, separate five-fold stability analysis, validation and synthetic testing. Subsequent reproducibility verification corroborated the principal recorded metrics and is documented in Section 3.7.3; repeated execution is not a new independent experiment or dataset. Actual record-level prediction evidence supports the figures, tables and paired comparison. The separately evaluated word–character and calibrated candidates are development-validation diagnostics and are distinguished from the original reference SVM. No authentic-data generalization, operational HITL improvement or application deployment is claimed.

4.2 Dataset Integrity and Similarity Diagnostics

The unchanged corpus contains 8,000 examples per class and 6,000 per assigned language group. Recorded partitions remain 19,213 training, 2,399 validation and 2,388 previously evaluated synthetic test records. Raw and Unicode-normalized exact duplicate excess are both zero. Record IDs are unique, and scenario-family, template-family and normalized-text overlaps between stored partitions are zero. These checks establish identifier and exact-text separation, not independent linguistic construction. A training-fitted reference unigram TF–IDF similarity diagnostic found nearest-training cosine similarity of at least 0.90 for 57.52% of validation narratives and 58.00% of test narratives. Median similarities were 0.911758 and 0.912752, respectively. This descriptive lexical threshold does not adjudicate duplicates. Shared phrases and class-conditional generator rules remain plausible explanations of ceiling scores. All language and human-label review flags remain negative.

4.3 Hyperparameter Search and Grouped Cross-Validation

Figure 14 presents all 18 recorded SVM configurations using training-only three-fold grouped Macro-F1. Every configuration scored 1.000000 with zero recorded fold dispersion; deterministic tie handling selected $C = 0.1$, word unigrams and minimum document frequency 2. The uniform grid cannot demonstrate that any parameter choice is intrinsically superior.

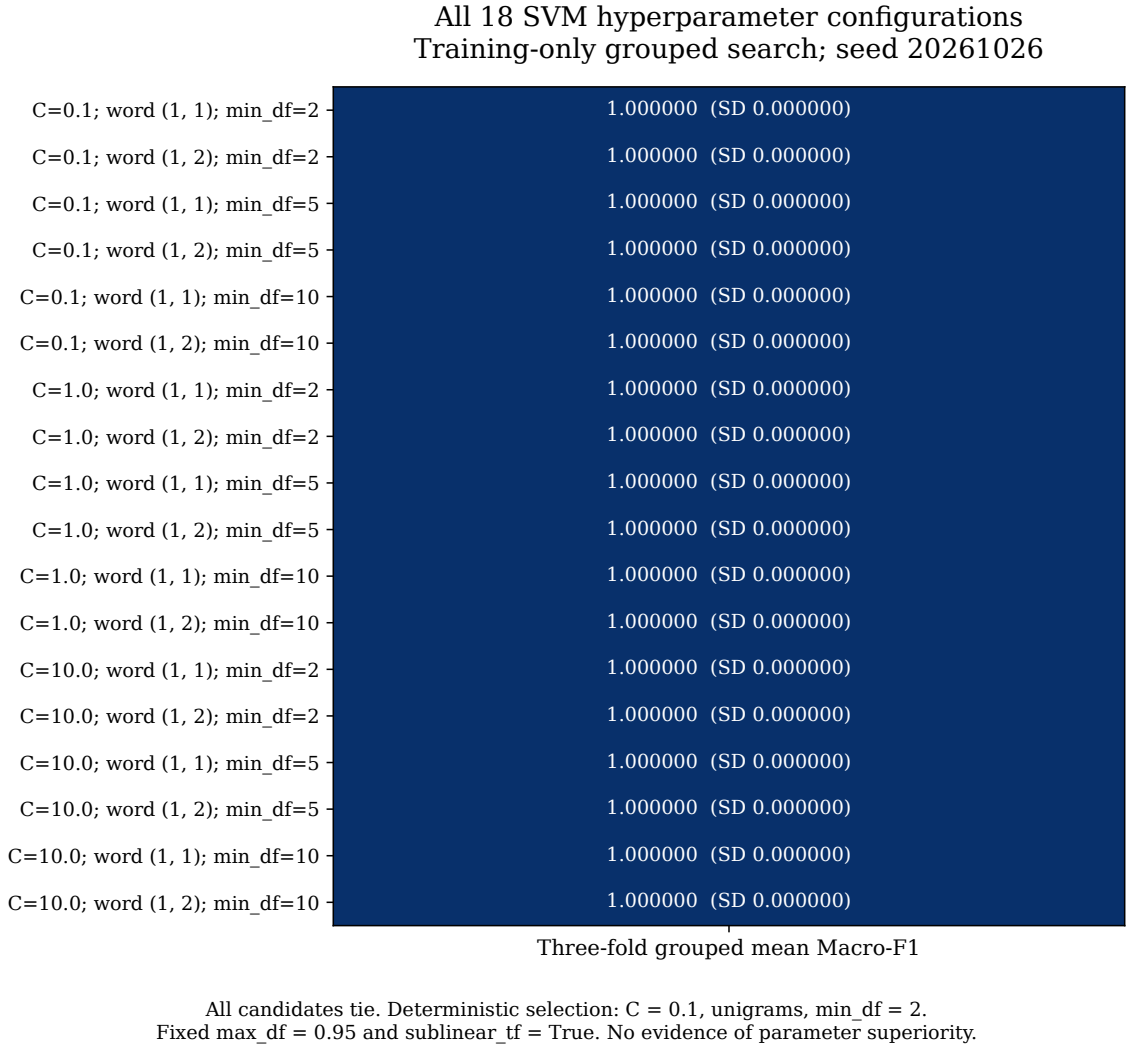
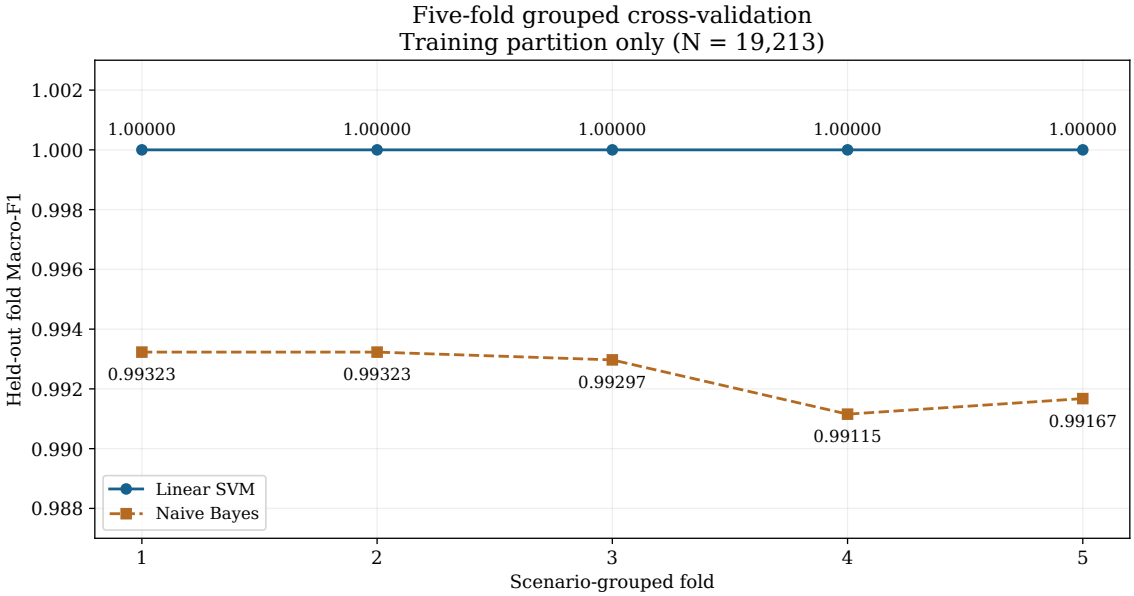


Figure 14: Three-fold grouped hyperparameter search over all 18 SVM configurations. Recorded values are supported by preserved outputs and reproducibility verification.

The separate five-fold stability analysis refitted the complete vectorizer/classifier pipeline within each training fold. It excluded the fixed validation and historical test records and verified zero scenario-family overlap within all folds. Figure 15 reports fold values and aggregate sample standard deviations. SVM mean Macro-F1 was 1.000000 (SD

0.000000), while Naive Bayes mean Macro-F1 was 0.992452 (sample SD 0.000971). This dispersion is descriptive and is not a population confidence interval or an independent nested model-selection estimate.



Pipeline	F1	F2	F3	F4	F5	Mean	Sample SD
Linear SVM	1.00000	1.00000	1.00000	1.00000	1.00000	1.000000	0.000000
Naive Bayes	0.99323	0.99323	0.99297	0.99115	0.99167	0.992452	0.000971

TF-IDF refitted inside every fold. SD describes fold dispersion, not a population confidence interval.

Figure 15: Five-fold scenario-grouped training-only cross-validation, including fold-level values, means and sample standard deviations.

4.4 Fixed Development Validation

The same two frozen reference pipelines were evaluated on 2,399 validation narratives. Linear SVM achieved 1.000000 Macro-F1 with 0 errors; Naive Bayes achieved 0.992077 Macro-F1 with 19 errors. Figure 16 gives accuracy, macro-precision, macro-recall, Macro-F1, Weighted-F1 and record counts. The validation partition is an existing development benchmark.

Fixed development validation performance

Metric	Linear SVM	Naive Bayes
Evaluated records	2399	2399
Accuracy	1.000000	0.992080
Macro-precision	1.000000	0.992267
Macro-recall	1.000000	0.992073
Macro-F1	1.000000	0.992077
Weighted-F1	1.000000	0.992079
Misclassifications	0	19

Precision and recall are macro-averaged; scores on a 0-1 scale.
Same fixed corpus; no independent field-validation claim.

Figure 16: Fixed Development Validation summary supported by the original results and verified reference predictions. Precision and recall are macro-averaged.

No errors occurred in the reference SVM on this within-generator partition. Figure 17 gives the actual counts, with reference labels on rows and predictions on columns.

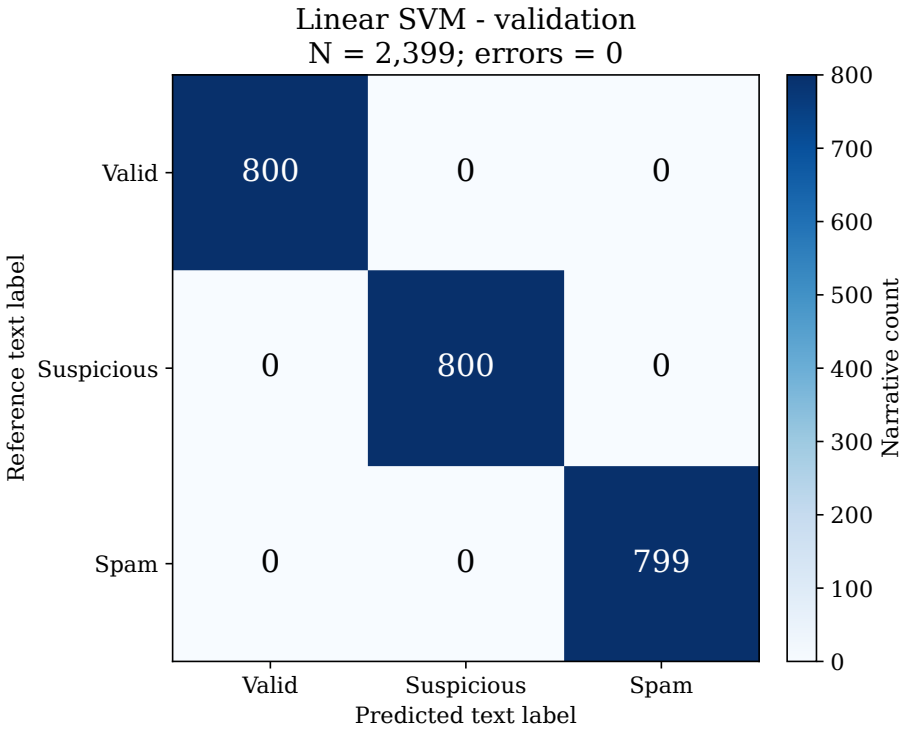


Figure 17: Linear SVM confusion matrix on the fixed development validation partition. Entries are narrative counts.

All 19 Naive Bayes errors were reference Spam narratives predicted as Suspicious. Figure 18 gives the actual counts, with reference labels on rows and predictions on columns.

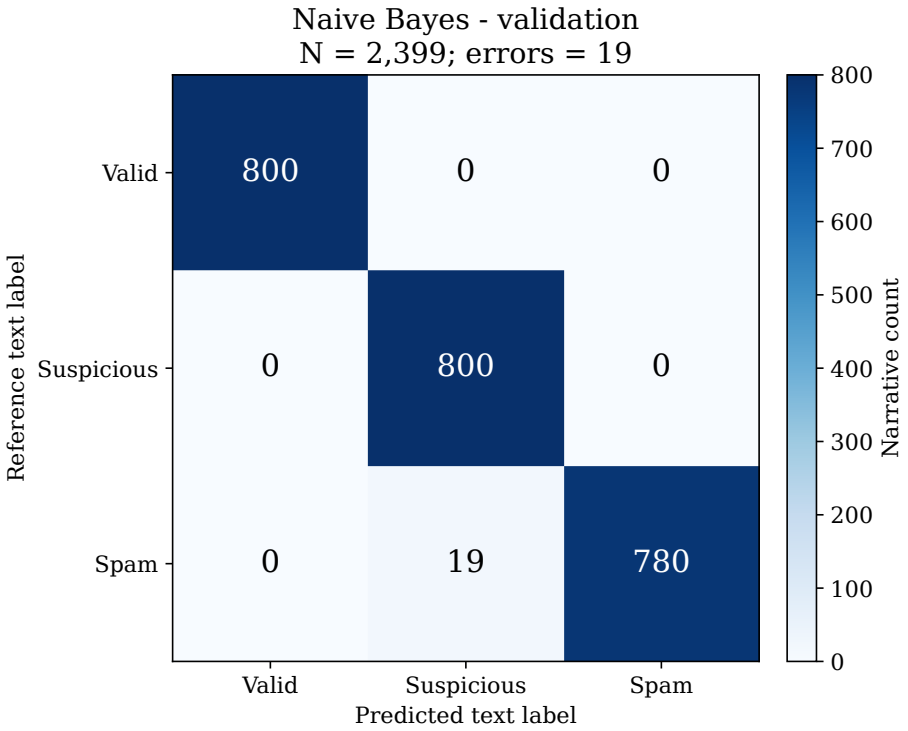


Figure 18: Naive Bayes confusion matrix on the fixed development validation partition. Entries are narrative counts.

Table 12: Per-class metrics on the fixed development validation partition

Pipeline	Class	Precision	Recall	F1	Support
SVM	Valid	1.000000	1.000000	1.000000	800
SVM	Suspicious	1.000000	1.000000	1.000000	800
SVM	Spam	1.000000	1.000000	1.000000	799
Naive Bayes	Valid	1.000000	1.000000	1.000000	800
Naive Bayes	Suspicious	0.976801	1.000000	0.988264	800
Naive Bayes	Spam	1.000000	0.976220	0.987967	799

4.5 Previously Evaluated Synthetic Test

The same two frozen reference pipelines were evaluated on 2,388 historically evaluated synthetic test narratives. Linear SVM achieved 1.000000 Macro-F1 with 0 errors; Naive Bayes achieved 0.991624 Macro-F1 with 20 errors. Figure 19 gives accuracy, macro-precision, macro-recall, Macro-F1, Weighted-F1 and record counts. These synthetic test results concern the original, already inspected test partition; subsequent verification does not make it an independent new test. It was not used for candidate feature or calibration selection.

Previously evaluated synthetic test performance

Metric	Linear SVM	Naive Bayes
Evaluated records	2388	2388
Accuracy	1.000000	0.991625
Macro-precision	1.000000	0.991780
Macro-recall	1.000000	0.991677
Macro-F1	1.000000	0.991624
Weighted-F1	1.000000	0.991624
Misclassifications	0	20

Precision and recall are macro-averaged; scores on a 0-1 scale.
Same fixed corpus; no independent field-validation claim.

Figure 19: Previously Evaluated Synthetic Test summary supported by the original results and verified reference predictions. Precision and recall are macro-averaged.

No errors occurred in the reference SVM on this within-generator partition. Figure 20 gives the actual counts, with reference labels on rows and predictions on columns.

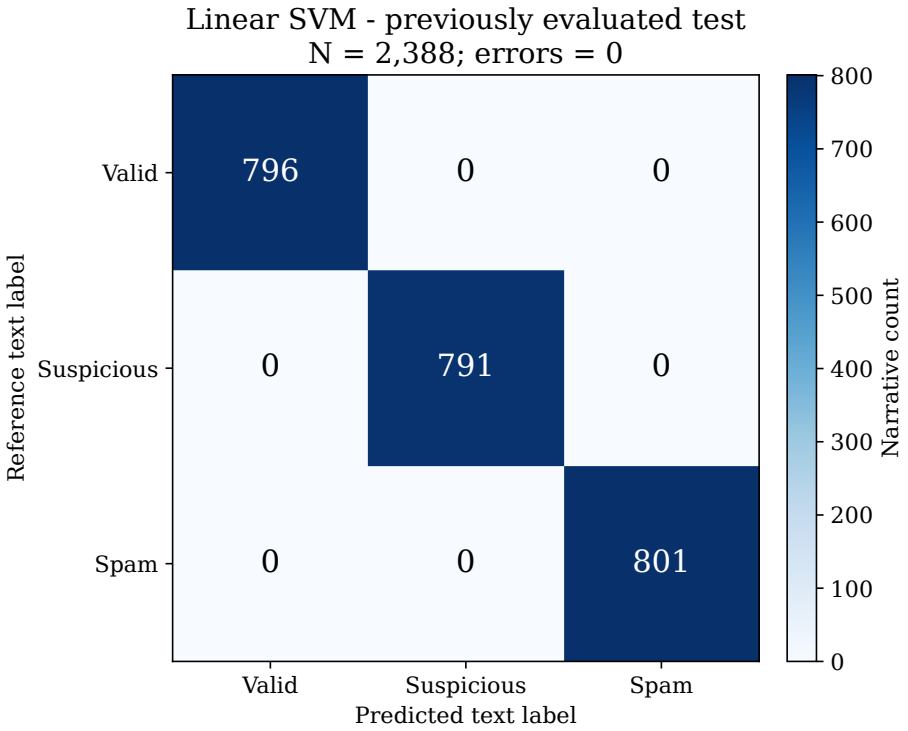


Figure 20: Linear SVM confusion matrix on the previously evaluated synthetic test partition. Entries are narrative counts.

All 20 Naive Bayes errors were reference Spam narratives predicted as Suspicious. Figure 21 gives the actual counts, with reference labels on rows and predictions on columns.

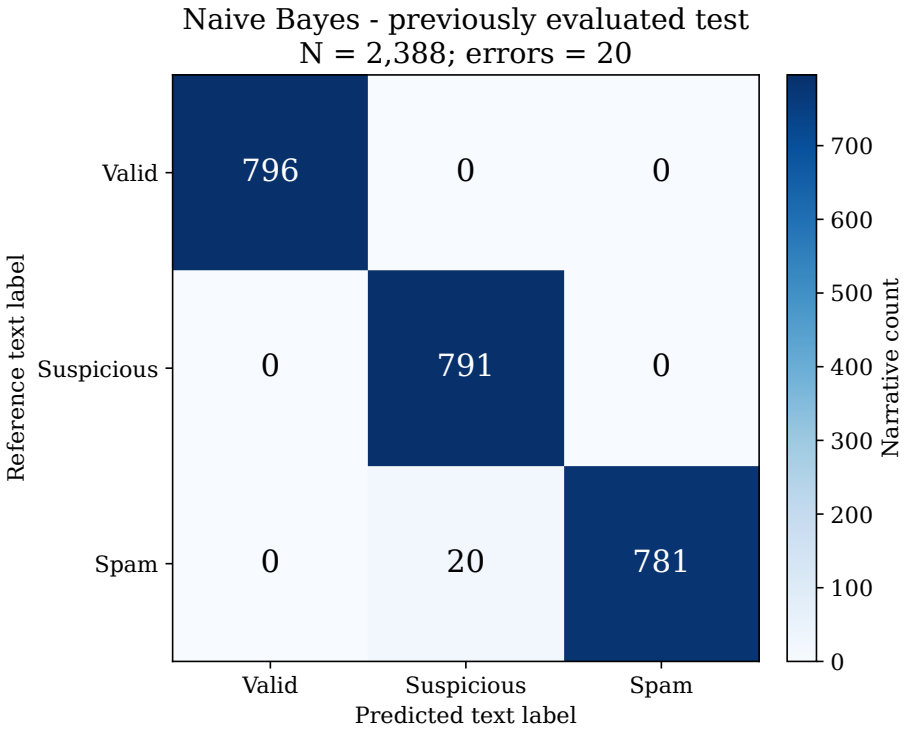


Figure 21: Naive Bayes confusion matrix on the previously evaluated synthetic test partition. Entries are narrative counts.

Table 13: Per-class metrics on the previously evaluated synthetic test partition

Pipeline	Class	Precision	Recall	F1	Support
SVM	Valid	1.000000	1.000000	1.000000	796
SVM	Suspicious	1.000000	1.000000	1.000000	791
SVM	Spam	1.000000	1.000000	1.000000	801
Naive Bayes	Valid	1.000000	1.000000	1.000000	796
Naive Bayes	Suspicious	0.975339	1.000000	0.987516	791
Naive Bayes	Spam	1.000000	0.975031	0.987358	801

4.6 Reference Pipeline Comparison

Figure 22 compares both reference pipelines across all five aggregate metrics and both partitions. Its full 0–1 axis prevents a visually exaggerated difference near the score ceiling. Hatching and distinct labels preserve readability in grayscale. The selected SVM uses word unigrams while the Naive Bayes baseline uses unigrams/bigrams; this is a comparison of complete pipelines, not a controlled isolation of estimator choice.

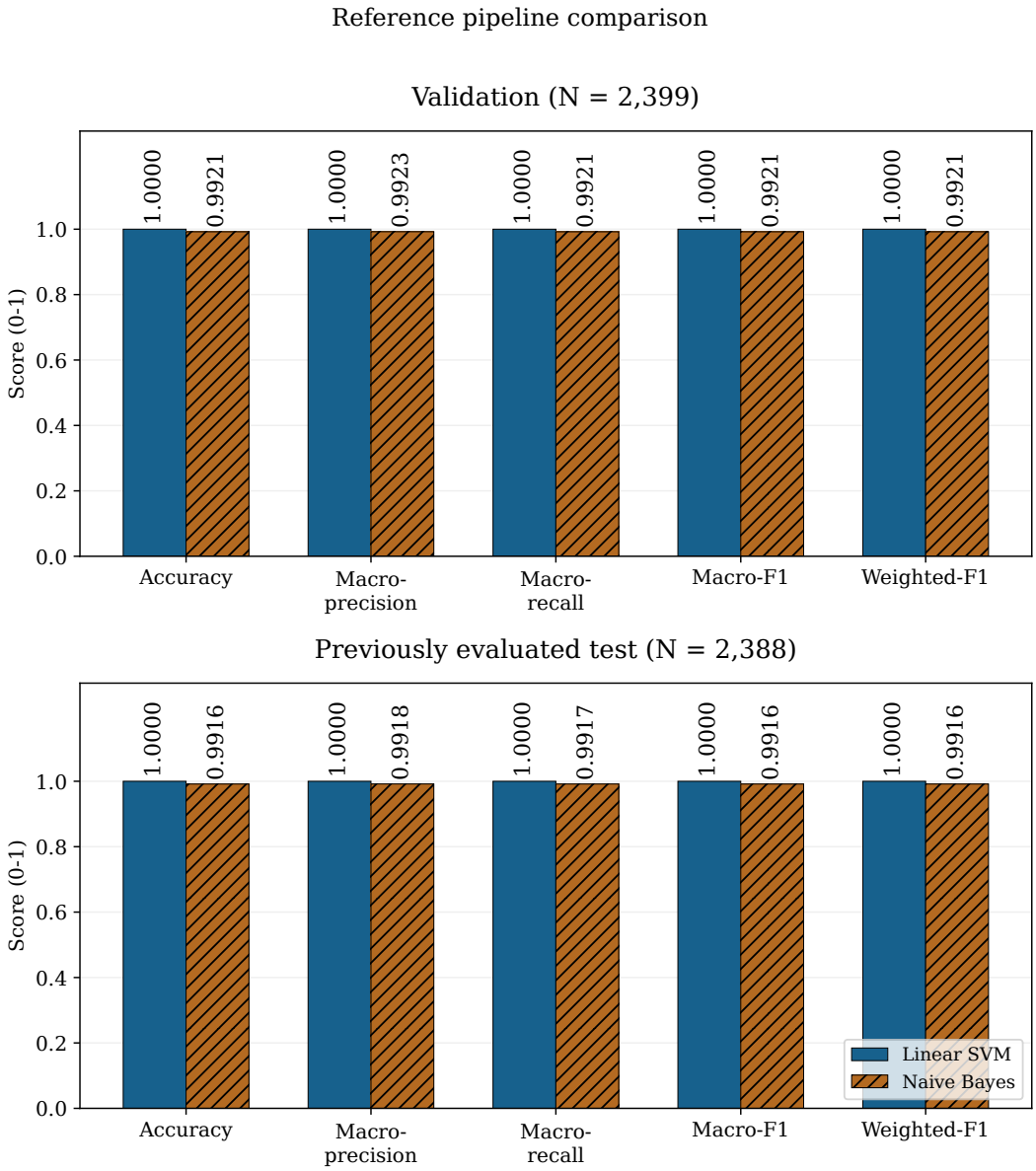


Figure 22: Reference SVM and Naive Bayes comparison on fixed validation and the previously evaluated synthetic test.

4.7 Paired Statistical Comparison

The reference predictions were exported with record identifiers and aligned by identifier and reference label before computing McNemar's test. Figure 23 presents both contingency tables, continuity-corrected statistics and exact two-sided binomial p-values. Validation has $b = 0, c = 19$, corrected $\chi^2 = 17.052632$ and $p = 3.814697266 \times 10^{-6}$. The synthetic test comparison has $b = 0, c = 20$, corrected $\chi^2 = 18.05$ and $p = 1.907348633 \times 10^{-6}$.

Paired McNemar comparison from record-level predictions
Validation (N = 2,399)

Baseline / SVM	SVM correct	SVM wrong
Naive Bayes correct	2380	0
Naive Bayes wrong	19	0

b = 0; c = 19; corrected chi-square = 17.052632
Exact two-sided p = 3.814697266e-06

Previously evaluated synthetic test (N = 2,388)

Baseline / SVM	SVM correct	SVM wrong
Naive Bayes correct	2368	0
Naive Bayes wrong	20	0

b = 0; c = 20; corrected chi-square = 18.050000
Exact two-sided p = 1.907348633e-06

Nominal p < 0.05 favors SVM on this synthetic benchmark.
Shared generator/family dependence limits inference; this is not evidence of field superiority.
Corrected statistic: max(0, |b-c|-1)^2 / (b+c); exact test: Binomial(b+c, 0.5).

Figure 23: Paired McNemar analysis computed from actual per-record reference predictions on both existing partitions.

Both nominal p-values are below 0.05 and favor the SVM's error incidence on this synthetic benchmark. This is not proof of field superiority, an effect of HITL, or equivalence of the underlying feature representations. Generated narratives share families and construction rules, which limits the independent-pair assumption and may make nominal p-values anticonservative. No probability of successful real-world complaint resolution follows from these tests.

4.8 Assigned-Language Performance

Language groups are generator metadata, not classifier inputs or independently verified language labels. Figure 24 reports validation subgroup support and Macro-F1. Each

SVM subgroup reaches the same ceiling while baseline errors vary. These findings cannot establish linguistic fairness or transfer to naturally authored code-switching.

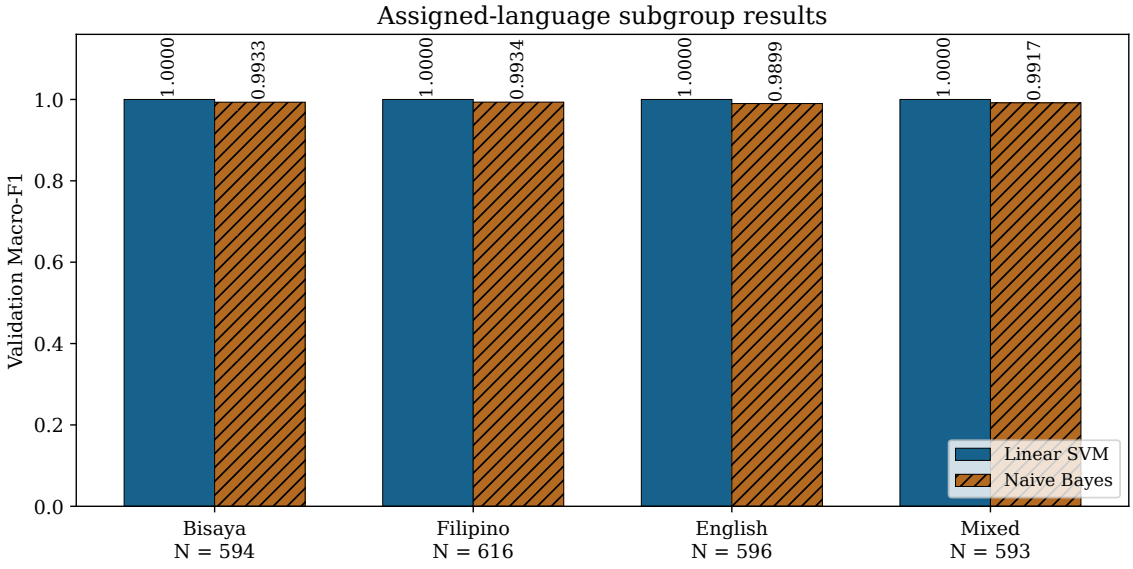


Figure 24: Validation performance by assigned language group; group supports are shown explicitly.

Table 14: Previously evaluated synthetic test by assigned language group

Language	N	SVM Macro-F1	NB Macro-F1
Bisaya	595	1.000000	0.989841
Filipino	591	1.000000	0.991517
English	603	1.000000	0.991748
Mixed	599	1.000000	0.993350

4.9 Learning and Shortcut Diagnostics

The original training-size procedure used the recorded fixed fractions. Validation Macro-F1 was 0.998749 at 1,921 training records and reached 1.000000 from the 20% fraction onward (Figure 25). Early saturation is consistent with easily learned generator patterns and does not establish sample sufficiency for authentic reports.

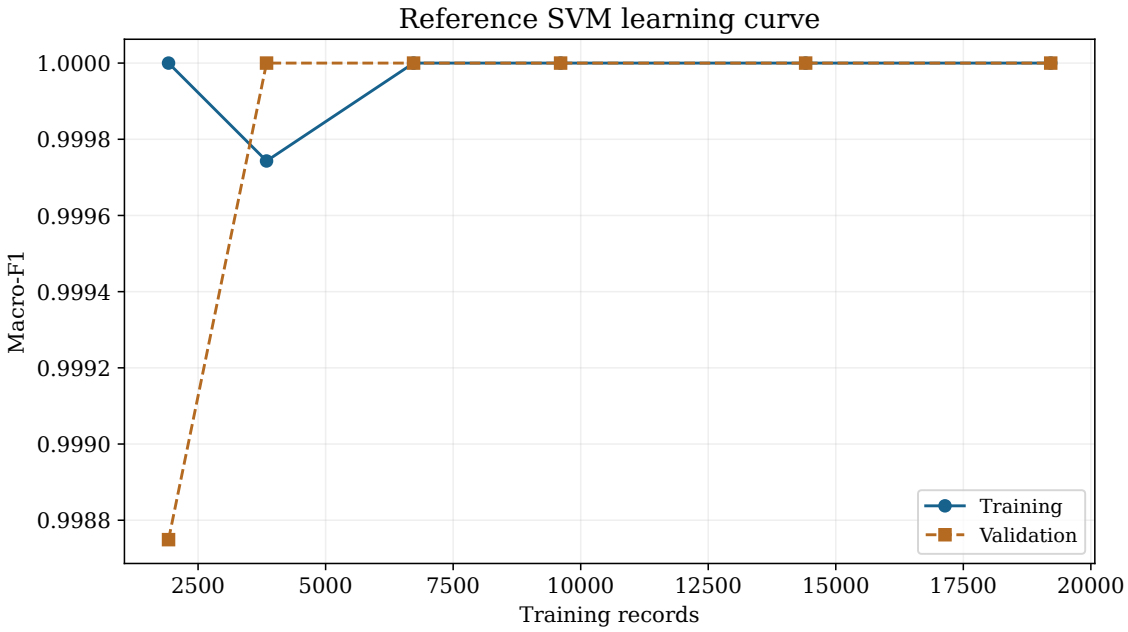


Figure 25: SVM training-size learning curve; sample count is not an optimizer epoch.

A separate SGD hinge-loss classifier supplies the epoch diagnostic in Figure 26; it is not the selected LinearSVC optimizer. Training hinge loss changed from 0.272574 to 0.009421, and validation loss from 0.300816 to 0.012279 over 15 passes. A decreasing auxiliary loss does not rule out generator-specific overfitting.

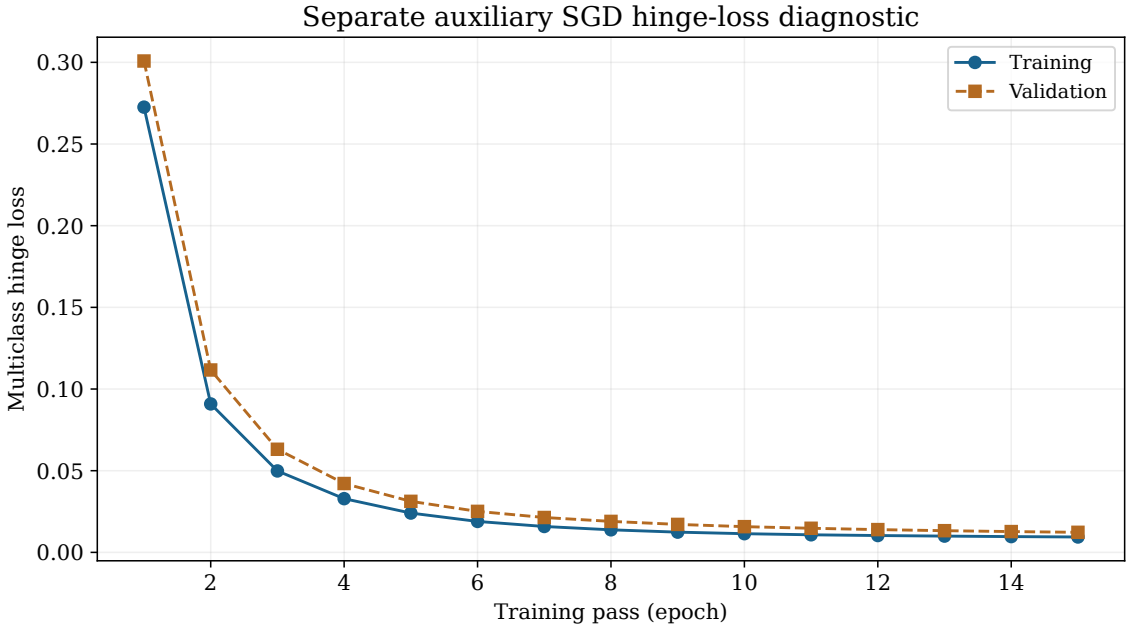


Figure 26: Measured auxiliary SGD training/validation hinge loss, explicitly separate from the final LinearSVC estimator.

The two-feature length-only baseline achieved 0.549396 validation accuracy and 0.537161 Macro-F1. This above-chance diagnostic confirms that narrative length carries class information. A separate character-only diagnostic also reached the synthetic validation ceiling. These are diagnostics, not evidence of semantic understanding. No independent challenge dataset was executed: the archived 96-message challenge results remain historical, while the optional prior 48-message input expected by the original script was absent.

4.10 Separately Evaluated Combined-Feature Candidate

In a separate development evaluation, one fixed candidate combined word unigrams/bigrams with character-within-word 3–5-grams, using conservative normalization and $C = 0.1$. A second version used sigmoid calibration through three scenario-grouped training folds. Both candidates were evaluated only on the 2,399 existing validation records; neither was run on the historical test. Figure 27 separates their development results from the reference measurements.

Separate development candidate experiment (N = 2,399)

Pipeline	Macro-F1	Errors	Log loss	ECE (10 bins)
Reference word SVM	1.000000	0	Not calibrated	Not calibrated
Word + character SVM	1.000000	0	Not calibrated	Not calibrated
Calibrated word + character	1.000000	0	0.00079512	0.00079433

One fixed candidate; sigmoid calibration uses three grouped training folds.
All classification scores tie; probability diagnostics are confined to synthetic development data.
Candidate models were not evaluated on the historical test partition.

Figure 27: Separately executed word-plus-character candidates, with and without grouped sigmoid calibration; development validation only.

Both candidates achieved 1.000000 validation Macro-F1 and zero errors, tying the word reference; therefore no classification improvement was demonstrated. The calibrated candidate yielded log loss 0.00079512, multiclass Brier sum 0.00000267, and ten-bin expected calibration error 0.00079433. These probability diagnostics are measured on the same synthetic development distribution and do not certify operational reliability. Candidate calibration is now an executed research component; probability-based live decisions and field calibration validation remain proposed.

4.11 Interpretation and Limits

The recorded metrics are supported by preserved data, group-aware feature fitting and subsequent reproducibility checks. Perfect SVM scores must be reported faithfully, but their credibility is bounded by synthetic, unreviewed labels, repeated construction patterns, high nearest-training similarity and an already inspected test set. No new authentic data, annotation agreement study, prospective feedback trial, web-system acceptance test or complaint-resolution effectiveness study was conducted. Next steps are independent annotation and external validation, followed by the separately specified application, HITL, privacy, case-management and model-release tests.

Chapter 5

Conclusions and Recommendations

5.1 Summary of Research Findings

The study documented and experimentally evaluated a text-only complaint-classification component using a controlled synthetic dataset. The 24,000-record corpus was deliberately balanced across three labels and four assigned language groups, with 118 researcher-defined generation categories used for systematic scenario coverage. The model design uses TF–IDF with Linear SVM and a Multinomial Naive Bayes baseline, and separates scenario metadata from the predictive narrative. The original recorded findings, corroborated by reproducibility verification, are: SVM validation/test Macro-F1 remained 1.000000 and baseline Macro-F1 was 0.992077/0.991624. A separate combined-feature candidate tied SVM on development validation; no classification improvement was demonstrated. The group-disjoint cross-validation and fixed validation results are high, although the exact ceiling effects, class-dependent text lengths, and features shared by generation patterns constrain the inference that can be drawn about unfamiliar citizen language. The work offers a reproducible research benchmark and a proposed human-governed complaint processing architecture; it does not establish municipality-wide predictive reliability or measured benefits of live feedback-driven retraining.

5.2 Recommendations and Future Work

5.2.1 Data Coverage and Representation Bias

The 118-category inventory and the synthetic text-generation rules should be audited with fluent Cebuano, Filipino and code-switched language reviewers. The current 8,000/8,000/8,000 class allocation is a controlled experimental design and should not be treated as actual submission prevalence. Future evaluations should include deliberately different reporting-channel styles, rare but meaningful complaints, safety-sensitive reports, varied lengths, misspellings and independent root scenarios. If authorized administrative reports are obtained later, they must be separated from synthetic scenarios by provenance and privacy controls; their collection is not required to state the current experimental findings accurately.

5.2.2 Independent Annotation and Feedback Bias

The presently generated labels have not received measured independent double annotation. A blinded sample spanning the full range of classifications, languages and difficult cases should be reviewed by at least two annotators, with Cohen's kappa and adjudication records reported. Future Human-in-the-Loop correction logs may reflect the judgment of a narrow set of administrators and reporting channels. They should be audited for class prevalence, language, case severity and review consistency before their use in retraining. Model improvement must be tested on independent frozen evaluation examples rather than assumed from the existence of correction records.

5.2.3 Generalization, Model Selection and Operational Safety

Additional testing should use held-out generator root families, unseen complaint intents, minimal pairs changing only the critical context, and realistic class-base-rate scenarios. Regularization, cautious preprocessing, group-aware cross-validation and model versioning should be retained; class weighting or training-only oversampling may be tested *if* feedback data later show meaningful imbalance. Neither SMOTE nor neural dropout is part of the current fitted TF–IDF + linear SVM pipeline. To establish operational readiness, the application also requires authorization, privacy, emergency referral, database and state-transition integrity, administrator audit trails, latency, monitoring and rollback testing. Safety-critical reports must remain accessible to human review even when a text classifier makes an error.

5.2.4 Accountable Case Management and Scope Boundaries

The proposed accepted-case workflow separates Unresolved, Ongoing and Resolved administrative statuses from Valid, Suspicious and Spam model predictions. It recommends a case tracking reference, responsible officer assignment, activity log, policy-specific response target, supervisor notification of overdue events, referral tracking, evidence review before closure and resident reconsideration. These are auditable system-design requirements rather than measured improvements in governmental response or verified legal compliance. Response duties under Republic Act No. 6713 and service-specific processing standards under Republic Act No. 11032 must be interpreted through the applicable Citizen's Charter; no universal deadline for complete incident resolution is assumed [29, 30, 7].

Optional image attachments, image classification, multimodal fusion, GPS maps and automated authenticity detection are deliberately deferred. A supporting location description can be requested where necessary without requiring GPS. The text-only SVM is not capable of independently deciding whether a physical incident occurred, whether an image is authentic, or whether a complaint was resolved.

5.2.5 Evaluation of Proposed Extensions

The combined word/character feature candidate and grouped sigmoid calibration have now been evaluated on existing synthetic development data only, tying the reference classifier without demonstrated classification improvement. Before field deployment, independently

validate transfer and probability reliability and implement/test the class-based review queue; conduct a double-annotation pilot and a representative audit of predicted Valid reports; evaluate a complete case-management prototype; and compare cumulative verified-feedback checkpoints of 100, 250 and 500 only if authentic, quality-checked labels are actually collected. These are experimental checkpoints, not guaranteed improvement thresholds [39, 41, 34]. No prospective feedback-checkpoint or operational HITL results are asserted in the present manuscript.

5.3 Concluding Statement

Controlled synthetic examples make it possible to vary complaint-writing attributes and expose classification weaknesses systematically, while documented EDA and evaluation protect against ungrounded claims. The contribution of the present work is the reproducible investigation of a multilingual three-label complaint-classification task. Further independent human review and operational testing are required before stronger conclusions about deployment or feedback-driven learning can be justified.

Appendix A

Dataset Evidence Register and Category Coverage

The following codes are the scenario-category IDs associated with the dataset generator. Each category is a research *coverage control*; none is a fourth or higher-order target of the classifier. The full evidence matrix with category-level source identifiers and operational qualifications is supplied as `research_evidence/SOURCE_TO_CATEGORY_EVIDENCE_MATRIX_V2.csv`. The evidence matrix remains subject to domain-specialist confirmation for specialized jurisdictional topics.

Table 15: The 118 researcher-defined generation categories

Code	Category description
VAL-01	Waste Management
VAL-02	Environmental Pollution
VAL-03	Drainage and Sewerage
VAL-04	Flooding and Water Accumulation
VAL-05	Public Sanitation
VAL-06	Pest and Vector Control
VAL-07	Road Maintenance
VAL-08	Street Lighting
VAL-09	Water Supply
VAL-10	Electricity and Utility Concerns
VAL-11	Sidewalks and Pedestrian Facilities
VAL-12	Public Facilities Maintenance
VAL-13	Construction and Structural Hazards
VAL-14	Noise Disturbance
VAL-15	Public Disturbance
VAL-16	Neighborhood Disputes
VAL-17	Harassment and Threats
VAL-18	Theft and Property Incidents

Code	Category description
VAL-19	Community Security Concerns
VAL-20	Substance-Related Community Concerns
VAL-21	Trespassing and Property Boundary Concerns
VAL-22	Traffic Management
VAL-23	Parking Violations
VAL-24	Public Transportation Concerns
VAL-25	Road Safety
VAL-26	Stray Animals and Animal Control
VAL-27	Animal Welfare
VAL-28	Public Health and Disease Prevention
VAL-29	Child Protection
VAL-30	Women's Protection and Domestic Violence
VAL-31	Senior Citizen and PWD Accessibility
VAL-32	Social Welfare and Community Assistance
VAL-33	Barangay Services and Administration
VAL-34	Business and Commercial Nuisances
VAL-35	Disaster Preparedness and Emergency Hazards
VAL-36	Public Property and Community Space Misuse
VAL-37	Debt and Financial Disputes
VAL-38	Family and Household Disputes
VAL-39	Housing, Land Use, and Occupancy
VAL-40	Agricultural and Farming Concerns
VAL-41	Livestock and Poultry Concerns
VAL-42	Barangay Health Center Services
VAL-43	Daycare and Early Childhood Services
VAL-44	Relief Goods and Assistance Distribution
VAL-45	Barangay Ordinance Violations
VAL-46	Online Harassment and Digital Fraud Reports
VAL-47	Market and Vendor Concerns
VAL-48	Telecommunications Infrastructure
VAL-49	Barangay Justice and Mediation Services
VAL-50	Official Misconduct and Financial Irregularities
VAL-51	Sangguniang Kabataan (SK) and Youth Programs
VAL-52	Barangay Assembly and Public Participation
VAL-53	Fisheries, Coastal, and River Concerns
VAL-54	Livelihood and Cooperative Programs
VAL-55	Barangay Security and Response Services
VAL-56	Missing Persons and Welfare Checks
VAL-57	Discrimination and Unequal Treatment
VAL-58	Education and Scholarship Assistance
VAL-59	Consumer and Retail Complaints

Code	Category description
VAL-60	Cemetery, Burial, and Funeral Service Concerns
SUS-01	Unclear Complaint Issue
SUS-02	Incomplete Incident Description
SUS-03	Unclear Location
SUS-04	Unidentified Subject or Object
SUS-05	Missing Critical Incident Details
SUS-06	Truncated Complaint Narrative
SUS-07	Ambiguous Complaint Statement
SUS-08	Unclear References
SUS-09	Uncertain Incident Occurrence
SUS-10	Unclear Complaint Intention
SUS-11	Mixed or Disorganized Narratives
SUS-12	Unclear Incident Status
SUS-13	Contradictory Location Details
SUS-14	Contradictory Dates or Times
SUS-15	Conflicting Incident Descriptions
SUS-16	Contradictory Involved Parties
SUS-17	Conflicting Incident Status
SUS-18	Inconsistent Event Sequence
SUS-19	Vague Secondhand Report
SUS-20	Unclear Alleged Incident
SUS-21	Incomplete Emergency Information
SUS-22	Ambiguous Quoted Information
SUS-23	Relevant Complaint Mixed With Unrelated Text
SUS-24	Conflicting Copied Information
SUS-25	Missing Previous Complaint Context
SUS-26	Missing Referenced Evidence or Attachment
SUS-27	Unreadable or Corrupted Complaint Narrative
SUS-28	Conflicting Complaint Form Information
SPA-01	Product Advertisement
SPA-02	Service Advertisement
SPA-03	Discount and Sales Promotion
SPA-04	Affiliate and Referral Promotion
SPA-05	Unsolicited Job Recruitment
SPA-06	Social Media and Channel Promotion
SPA-07	Fake Prize and Lottery Notifications
SPA-08	Financial Institution Impersonation
SPA-09	Credential Phishing
SPA-10	OTP and Verification Code Scams
SPA-11	Advance-Fee and Payment Scams
SPA-12	Investment and Quick-Money Schemes

Code	Category description
SPA-13	Greeting-Only Messages
SPA-14	Casual Personal Conversation
SPA-15	Jokes and Meme Content
SPA-16	Entertainment and Celebrity Content
SPA-17	Unrelated Academic or Technical Requests
SPA-18	Unrelated Chain and Inspirational Messages
SPA-19	Random Keyboard Input
SPA-20	Nonsensical Word Combinations
SPA-21	Emoji and Symbol-Only Spam
SPA-22	Contextless Numbers and Codes
SPA-23	Placeholder and Test Messages
SPA-24	Link-Only Spam
SPA-25	Repetitive Promotional Messages
SPA-26	Automated Advertisement Templates
SPA-27	Mass-Forwarded Chain Messages
SPA-28	Keyword-Stuffing Spam
SPA-29	Classification Manipulation Attempts
SPA-30	Deliberate Off-Topic Trolling

Appendix B

Dataset Schema and Supporting Experimental Records

The released dataset contains `complaint_text`, `classification`, language-group allocation, category IDs, source references, scenario and template family identifiers, generation seed and split assignment. These additional fields support audits but must be *excluded* from the primary TF-IDF predictive features. The corpus is marked `synthetic_researcher_constructed_unreviewed`; generated-language and human-label validation flags remain "No". The archived run outputs document all directly reported numbers in Chapter 4. Further human reviews, prospective HITL retraining or production benchmark data are not included because no completed results were provided for those procedures.

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